

Natural variation in salt-induced changes in root:shoot ratio reveals SR3G as a negative regulator of root suberization and salt resilience in Arabidopsis

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eLife Assessment

Through cellular, developmental, and physiological analysis, this **valuable** study identifies a gene that regulates the relative growth of roots and shoots under salt stress. The holistic approach taken provides **solid** evidence that this member of a larger tandemly duplicated gene family together with an upstream regulator contributes to salt tolerance, although the statistical or biological support for some conclusions could be more robust. The manuscript will be of interest to plant biologists studying mechanisms of abiotic stress tolerance and gene family evolution.

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Abstract

Soil salinity is one of the major threats to agricultural productivity worldwide. Salt stress exposure alters root and shoot growth rates, thereby affecting overall plant performance. While past studies have extensively documented the effect of salt stress on root elongation and shoot development separately, here we take an innovative approach by examining the coordination of root and shoot growth under salt stress conditions. Utilizing a newly developed tool for quantifying the root:shoot ratio in agar-grown Arabidopsis seedlings, we found that salt stress results in a loss of coordination between root and shoot growth rates. We identify a specific gene cluster encoding domain-of-unknown-function 247 (DUF247), and characterize one of these genes as Salt Root:shoot Ratio Regulator Gene (SR3G). Further

analysis elucidates the role of *SR3G* as a negative regulator of salt stress tolerance, revealing its function in regulating shoot growth, root suberization, and sodium accumulation. We further characterize that *SR3G* expression is modulated by *WRKY75* transcription factor, known as a positive regulator of salt stress tolerance. Finally, we show that the salt stress sensitivity of *wrky75* mutant is completely diminished when it is combined with *sr3g* mutation. Together, our results demonstrate that utilizing root:shoot ratio as an architectural feature leads to the discovery of new stress resilience gene. The study's innovative approach and findings not only contribute to our understanding of plant stress tolerance mechanisms but also open new avenues for genetic and agronomic strategies to enhance crop environmental resilience.

Introduction

Salt stress is predominant in the dryland areas where evaporation rate exceeds water input. As all water contains dissolved ions, the prolonged exposure to drought stress results in increased accumulation of salts in the upper soil layers (Abrol et al., 1988 [\[1\]](#); Fischer et al., 2021 [\[2\]](#); Richards, 1969 [\[3\]](#)). Dryland areas of South America, southern and western Australia, Mexico, southwest United States, and South Africa were previously identified as primary hotspots for soil salinization (Hassani et al., 2021 [\[4\]](#)). As fresh water becomes scarcer, understanding the mechanisms of salt stress resilience becomes of paramount importance in the context of global agricultural productivity and sustainability.

Salt stress detrimentally affects plant growth and development, decreasing the activity of the meristem (West et al., 2004 [\[5\]](#)). Studies utilizing main root elongation as an indicator of salt tolerance have led to identification of many crucial components of salt stress signaling and tolerance, including genes involved in salt-overly sensitive signaling (SOS) pathway, jasmonate, ethylene, abscisic acid (ABA), and reactive oxygen species (ROS) in glycophytes and halophytes alike (Fu et al., 2019 [\[6\]](#); Geng et al., 2013 [\[7\]](#); Hu et al., 2023 [\[8\]](#); Şekerçi et al., 2023 [\[9\]](#); Valenzuela et al., 2016 [\[10\]](#); Wu et al., 1996 [\[11\]](#)). Additionally, negative halotropism response, that describes a directional growth of main root away from the salt gradient, was used to identify a wide range of molecular components regulating the auxin redistribution and cell-wall remodeling, to actively avoid high salt (Deolu-Ajayi et al., 2019 [\[12\]](#); Galvan-Ampudia et al., 2013 [\[13\]](#); Korver et al., 2020 [\[14\]](#); Yu et al., 2022 [\[15\]](#); Y. Zou et al., 2022 [\[16\]](#)). The development of lateral roots is affected by salinity, and the differential effect on main and lateral roots within specific genotype can result in substantial reprogramming of root system architecture (Julkowska et al., 2017 [\[17\]](#), 2014 [\[18\]](#)). The shoot development is impacted by salt stress, as the salt impacts the plant transpiration and photosynthetic efficiency instantaneously (Awlia et al., 2021 [\[19\]](#), 2016 [\[20\]](#); Julkowska et al., 2016 [\[21\]](#)). Many forward genetic studies were highly successful in associating natural variation in root and shoot growth with allelic variation in gene coding and promoter regions, thereby identifying potential new target traits for improved stress resilience (Awlia et al., 2021 [\[19\]](#); Julkowska et al., 2017 [\[17\]](#), 2016 [\[20\]](#)). However, these studies typically focus on either above or below ground tissue.

The coordination of root and shoot growth rate is an important aspect of seedling establishment, and becomes even more critical under abiotic stress (Bacher et al., 2022 [\[22\]](#); Chen et al., 2022 [\[23\]](#); Karcher et al., 2008 [\[24\]](#); Wang et al., 2018 [\[25\]](#); Xu et al., 2015 [\[26\]](#)). The importance of increased root:shoot ratio, maximizing water scavenging while limiting transpiration, is well studied under drought stress (Asch et al., 2005 [\[27\]](#); Chen et al., 2022 [\[23\]](#); Karcher et al., 2008 [\[24\]](#); Uga et al., 2013 [\[28\]](#)). However, the effect of salt stress on root:shoot ratio is yet to be described, and its contribution to salt stress remains unknown. Here, we developed a tool to quantify root:shoot ratio from agar-grown *Arabidopsis* seedlings, and re-analyzed the natural diversity panel of *Arabidopsis* exposed to salt stress (Julkowska et al., 2017 [\[17\]](#)). We have observed that while salt stress does not elicit a clear response in root:shoot ratio, it results in loss of coordination of root and shoot growth rates. The locus associated with salt-induced changes in root:shoot ratio consisted of a gene cluster

encoding domain-of-unknown-function 247 (DUF247), with majority of SNPs clustering into one specific gene (At3g50160), that we named as Salt Root:shoot Ratio Regulator Gene (SR3G). Further characterization of SR3G revealed its function as a negative regulator of salt stress resilience, through regulation of root suberization, shoot growth, and Na^+ accumulation. We found WRKY75 transcription factor, a positive regulator of salt stress tolerance, acts upstream of SR3G, binding directly to the SR3G promoter in yeast. While *sr3g* and *wrky75* mutants showed contrasting salt stress response, the double mutants of *sr3g/wrky75* displayed reduced sensitivity to salt stress.

Materials & Methods

Screening HapMap population for the increase in root and shoot size

The images collected within (Julkowska et al., 2017 [DOI](#)) were re-analyzed in this manuscript to examine salt-induced changes in the root:shoot ratio for the data presented here. 360 accessions of the Arabidopsis HapMap population (Weigel and Mott, 2009 [DOI](#)) were screened over 7 experimental batches, using 4 biological replicates per genotype per condition. The seeds were surface sterilized, incubated in sterile MQ water at 4°C for 48 h, and germinated on media containing ½ Murashige-Skoog, 0.5% sucrose, 0.1% MES buffer, pH 5.8 (KOH) and 1% Dashin agar. The 4-day-old seedlings were subsequently transferred to media additionally supplemented with 0, 75 or 125 mM NaCl. The plates were scanned every 2nd day using a flat-bed scanner for 8 consecutive days.

The collected images were analyzed for green and white pixels, corresponding to the projected surface of the shoot and root, respectively (<https://github.com/ronellsicat/PlantSeg> [DOI](#)). A Matlab-based tool was developed to simplify and speed up the segmentation and analysis pipeline. For automatic segmentation, the tool uses a combination of image operations (histogram equalization), thresholding on different color spaces (e.g., RGB, YCbCr, Lab, HSV), and binary image processing (boundary and islands removal). As the tool is digitalizing various color scales and classifies pixels into either white (root), green (shoot) or blue (background) categories, the adjustment for white balance is obsolete. For analysis, the tool uses connected components analysis to extract the plants (combined roots and shoots) and can guide the user to find possible mistakes which can be manually corrected via the user interface. After segmentation, the number of pixels for roots and shoots are counted and recorded automatically. More details are provided in the publicly available repository for the tool (<https://github.com/ronellsicat/PlantSeg> [DOI](#)). The collected images and corresponding data acquired from them using the tool can be accessed at Zenodo Repository (experimental batch 1: <https://doi.org/10.5281/zenodo.7272155> [DOI](#); experimental batch 2: <https://doi.org/10.5281/zenodo.7272344> [DOI](#); experimental batch 3: <https://doi.org/10.5281/zenodo.7272647> [DOI](#); experimental batch 4: <https://doi.org/10.5281/zenodo.7271841> [DOI](#); experimental batch 5: <https://doi.org/10.5281/zenodo.7272139> [DOI](#); experimental batch 6: <https://doi.org/10.5281/zenodo.7273936> [DOI](#); experimental batch 7: <https://doi.org/10.5281/zenodo.7273937> [DOI](#); quantified data: <https://doi.org/10.5281/zenodo.7268673> [DOI](#)).

The tool's precision was evaluated by growing the Col-0 seedlings on agar plates, analyzing their root and shoot projected area on the final day using the tool, and correlating it with the recorded fresh and dry weight of the seedlings (**Fig. S1**). The collected data from the HapMap panel was analyzed using R for potential outliers (samples identified to be more than 3x standard deviation from the mean of each genotype within each condition), and the genotype-specific mean was calculated using the stats package (Team et al., 2018 [DOI](#)). The overall increase in root and shoot area was investigated for differences between control and salt stress conditions (**Fig. S2**). The root and shoot growth were estimated by fitting a linear function for a log-transformed projected root or shoot area over the non-transformed timeline of the experiment (<https://rpubs.com/mjulkowska/HapMap-root-shoot-data> [DOI](#)). The data were subsequently used to explore differences between the

experimental batches and the standard deviations within individual accessions. The genotypes above 3 standard deviations from the population's mean were excluded from further analysis. The salt tolerance index (STI) was calculated for root and shoot growth by dividing the average trait value at salt stress by the genotype-specific average at control conditions (**Fig. S3**). The correlations between individual traits were examined using the genotypic mean (**Fig. S4**). All quantitative data for root and shoot growth was visualized using *ggplot2*, *corrplot* and *ggpubr* packages (Kassambara, n.d. [↗](#); Wei and Simko, 2017 [↗](#); Wilke, 2016 [↗](#)). For all measured traits, we estimated broad sense heritability (H^2) using the MVar (Julkowska et al., 2018 [↗](#)) (**Table S1**). The data used as an input for GWAS can be found in **Table S2**.

Genome-Wide Association Study

The genotype-specific mean was used as an input to GWAS. The GWAS was conducted as described in (Awlia et al., 2021 [↗](#)), initially using 250 k SNPs (Horton et al., 2012 [↗](#)) and subsequently using 4 M SNPs (1001genomes.org). The GWAS associations were evaluated for minor allele count (MAC) and association strength above the Bonferroni threshold with $-\log_{10}(\text{p-value}/\text{\#SNPs})$, calculated for each sub-population of SNPs above threshold MAC (**Table S3**, $\text{Bonf.threshold.MAC.specific}$). The GWAS data was processed for associations with all SNPs used to generate Manhattan and QQ-plots (R notebook for processing GWAS data: <https://rpubs.com/mjulkowska/GWAS-root-shoot-Arabidopsis-salt-250kSNPs> [↗](#)). Additionally, we prioritized the genetic loci where multiple SNPs were associated with individual traits, or multiple traits were associated with one genetic locus. Our final selection yielded 14 associations with traits measured under control conditions, while 9 and 23 associations were specific to traits measured under 75 and 125 mM NaCl, respectively (**Table S3**). The GWAS output files can be accessed at Zenodo Repository (GWAS using 250 k SNPs: <https://doi.org/10.5281/zenodo.7271751> [↗](#); GWAS using 4 M SNPs: <https://doi.org/10.5281/zenodo.7271741> [↗](#)). For the most promising loci (**Table S4**), the sequence information of 147 accessions belonging to the HapMap population was downloaded from the 1001 genomes project website (1001genomes.org) and aligned with ClustalO as described in (Julkowska et al., 2016 [↗](#)). More details on plotting the genome divergence in the loci of interest can be found at <https://www.protocols.io/view/plotdivergence-for-plotting-the-natural-variation-x54v98y84l3e/v1> [↗](#). The input and output files of Plot Divergence analysis can be found in Zenodo repository (<https://doi.org/10.5281/zenodo.7278021> [↗](#)).

Haplotype analysis

For the most promising candidate loci (**Table S4**), we have identified the gene open reading frames that were located within the genome-wide linkage-disequilibrium (LD) of the associated SNPs. The LD was expanded if multiple SNPs were identified within the region, and the region of interest was expanded based on the number of coding genes within the LD window. The protein-coding genes were subsequently used for haplotype analysis. All the SNPs located within the gene coding region were used to perform a haplotype analysis. The detailed protocol used for haplotype analysis can be accessed at <https://www.protocols.io/view/haplotype-analysis-of-the-cds-in-arabidopsis-thali-5jyl86n8v2wp/v1> [↗](#). The accessions were grouped into haplotype groups depending on the combination of SNPs within the protein-coding region of the gene of interest. The haplotypes represented by less than 3 accessions were excluded from further analysis. The phenotypes of individual haplotypes were explored for significant differences using ANOVA. The results for all investigated haplotypes can be accessed at Zenodo Repository (<https://doi.org/10.5281/zenodo.7277703> [↗](#)).

Identification of SR3G orthologs and paralogs

DUF247 domain-containing orthologs from *Arabidopsis thaliana* and eight other species were identified by reciprocal best BLAST within CoGe ((Nelson et al., 2018 [↗](#)); <https://genomevolution.org/CoGe/CoGeBlast.pl> [↗](#)) using SR3G (At3g50160) nucleotide sequence as query with an E-value cutoff of 1.00E-10). The gene ID, species, and gene names been used in this study are listed in **Table S5**.

Sequence homologs were further confirmed by multiple alignments of those hits within each species using Geneious Prime. Sequences with missing or incomplete DUF247 domains were removed, with the exception of the mis-annotated DUF427 homologs in *Camelina sativa* which contained an extraneous intron that, after removal, resulted in a set of intact DUF247 open reading frame containing genes.

Phylogenetic and positive selection analysis of SR3G orthologs

Identified orthologs were translated into amino acid, then aligned by MUSCLE. Alignments were carefully validated by manual correction. Briefly, diverged insertions were trimmed, shifted frames were adjusted, and conserved regions after translation were retained for phylogeny reconstruction. Phylogenetic reconstruction was performed using the RAXML (Stamatakis, 2014) plug-in within Geneious (Geneious Prime 2022.1.1) with the GAMMA-GTR nucleotide model, rapid bootstrapping with a parsimony search, and 100 bootstraps. The distribution of SR3G and other DUF247 orthologs, root lengths for each branch, were visualized by software Geneious Prime.

To characterize any potential positive selection that might exist along certain branches of the inferred phylogeny, including SR3G and its orthologs, we performed a branch-site model A test on branches leading to SR3G, and branches leading to the clade containing SR3G and its closet orthologs using PAML4.0 (Stamatakis, 2014; Yang, 2007) as performed previously (Beilstein et al., 2015; Nelson et al., 2014). Specific branches are denoted as ω_0 , ω_1 , ω_2 , ω_3 , and ω -rest (Table S6). In the null hypothesis (H_0): we assigned a substitution rate ($\omega = dN/dS$) along with $0 < \omega < 1$ or $\omega = 1$ to all branches (represented by H_0 : $\omega_0 = \omega_1 = \omega_2 = \omega_3 = \omega$ -rest). The alternative hypotheses (HAs) here allow for differential selection to be applied to respective branch(es). Specifically, the five alternative hypotheses are listed here: H_1 : ω -rest branches = $\omega_0 = \omega_3 = \omega_1 \neq \omega_2$, H_2 : ω -rest branches = $\omega_0 = \omega_1 = \omega_2 \neq \omega_3$, H_3 : ω -rest branches = $\omega_0 \neq \omega_1 \neq \omega_2 \neq \omega_3$, H_4 : ω -rest branches $\neq \omega_0 \neq \omega_1 \neq \omega_2 \neq \omega_3$, H_{sep} : each branch has its unique ω value. To test these hypotheses, we further validate or reject alternative hypotheses by conducting a Chi-square test based on likelihood scores (Ln-L score) derived from the respective branch-site model A tests (Chi-square test cutoff: $P < 0.05$).

Screening candidate genes identified through GWAS

The genes located within the selected loci (Table S4) were further inspected for their contributions to root development and root:shoot ratio under control and salt stress conditions. The T-DNA insertion lines (Table S4) were genotyped using primers listed in Table S4, to confirm the homozygous insertion of the T-DNA within the phenotype lines. All the mutant lines were used for seed propagation in the controlled growth chamber with 16/8h of light/dark period, at 22°C, 60% humidity and 200 $\mu\text{mole m}^{-1} \text{s}^{-1}$ under long-day conditions. The seeds of plants identified as homozygous were subsequently used for the experiment on agar plates, as described before for the Genome-Wide Association Study. A detailed protocol for plate assays can be accessed at <https://www.protocols.io/view/plate-assay-for-identification-of-salt-stress-indu-eq2lyd1mwlx9/v1>. The plate images were analyzed for root system architecture using Smart Root (Lobet et al., 2011) and the root:shoot ratio using the tool described before for Genome-Wide Association Study. The analysis of the mutant lines was performed in R. The data was first curated for selecting possible outliers within a genotype and subsequently used to fit growth functions for root and shoot tissues using exponential functions.

Screening SR3G mutants within root:shoot locus

The following T-DNA lines (listed in Table S4) were used for SR3G (AT3G50160) as the main GWAS candidate: *sr3g-4* (SAIL_690_E12, glufosinate resistant) and *sr3g-5* (SAIL_608_C06, glufosinate resistant). All T-DNA insertion alleles were in the background of Col-0 accession. The position of T-DNA insertions was verified using PCR (primers listed in Table S4). RNAi lines targeting the neighboring of the SR3G or a combination of SR3G and its neighboring DUF247 genes included

CSHL#3 targeting DUF130, CSHL#4 targeting SR3G, CSHL#5 targeting DUF130 and DUF170, CSHL#6 targeting DUF130 and DUF140, CSHL#7 targeting DUF10 and DUF150, CSHL#8 targeting DUF140 and DUF190, CSHL#10 targeting DUF180, CSHL#11 targeting DUF150 and DUF170, CSHL#12 targeting DUF130, DUF140, DUF170, DUF180, and DUF190, CSHL#13 targeting DUF140 and DUF170, CSHL#15 targeting DUF140, DUF170, DUF180, and DUF190, CSHL#16 targeting DUF130, SR3G, and DUF170, CSHL#18 targeting DUF150 and DUF190, CSHL#19 targeting DUF130, DUF140, DUF150, SR3G, DUF170, DUF180, and DUF190, CSHL#20 targeting DUF130 and SR3G, CSHL#21 targeting DUF130 and DUF150, and CSHL#22 targeting DUF150 and SR3G. The RNAi lines were verified using RT-qPCR, with the primers listed in **Table S4**. The following T-DNA insertion alleles (also listed in **Table S4**) were used to characterize the neighboring genes of the *SR3G* (AT3G50160): 120.2 (SAIL_382_A09), 140.1 (SALK_044685), 140.2 (SALK_005466), 140.3 (SALK_122700), 150.1 (SALK_003824), 150.3 (SALK_071080), 170.1 (SALK_009186), 170.2 (SALK_112602), 170.3 (SALK_008710), 170.4 (SALK_145999), 170.5 (SALK_072937), 190.2 (SALK_137791), 200.1 (SALK_113759), 200.2 (SALK_093701), 200.3 (SALK_129634). The primers used for genotyping PCR are shown in **Table S4**.

Screening WRKY75 mutant lines and generation of *wrky75/sr3g* double mutant

The following T-DNA insertion alleles were used for *WRKY75* (AT5G13080): *wrky75-1* (SALK_004954, kanamycin resistant) and *wrky75-3* (SALKseq_046273, kanamycin resistant). All T-DNA insertion alleles were in the background of Col-0 accession. The primers used for genotyping PCR are shown in **Table S4**. Two sets of double mutant combinations were generated for *wrky75-3/sr3g-5*, with *wrky75-3* serving as the maternal parent in one (referred to as line d) and *sr3g-5* serving as the maternal parent in the other (referred to as line e).

Plasmid construction and plant transformation

For plant expression, all clones were constructed using the GreenGate cloning system (Lampropoulos et al., 2013). We have generated full-length *SR3G* constructs, as well as three fragments – fragment A (from amino acid number 1 to 216), fragment B (from amino acid number 217 to 468) and fragment C (from amino acid number 469 to 503). Fragments A, B, AB, and ABC (full length) were cloned into entry vector pGGC (GreenGate) with mVenus as N-terminal tag. *SR3G* fragments were PCR amplified using high-fidelity DNA polymerase, and cloned as full-length cDNA or truncated version. The entry clone was evaluated using Sanger sequencing to verify the absence of mutations. Promoters used included Ubiquitin 10 promoter (Norris et al., 1993). The list of constructs generated within this project can be found in **Table S7**. The *SR3G* constructs generated in this paper and their corresponding vector maps can be accessed in Zenodo repository ([https://zenodo.org/records/10788996](https://zenodo.org/records/10788996?token=eyJhbGciOiJIUzUxMiJ9.eyJpZCI6IjI2ZjkzZGRlLTQ3M2YtNDNmZS04MmM5LTNhMjQ5MGFmOWY0NiIsIHBxOSBIH2f8txZObh5YRWH4Hkva0pzqXQ)) ([token=eyJhbGciOiJIUzUxMiJ9.eyJpZCI6IjI2ZjkzZGRlLTQ3M2YtNDNmZS04MmM5LTNhMjQ5MGFmOWY0NiIsIHBxOSBIH2f8txZObh5YRWH4Hkva0pzqXQ](https://zenodo.org/records/10788996?token=eyJhbGciOiJIUzUxMiJ9.eyJpZCI6IjI2ZjkzZGRlLTQ3M2YtNDNmZS04MmM5LTNhMjQ5MGFmOWY0NiIsIHBxOSBIH2f8txZObh5YRWH4Hkva0pzqXQ)). The final constructs were validated via vector digestion, using BaeGI for the constructs with fragments A, B, and AB and PvuI for the full length *SR3G* construct. Subsequently, we transformed them into *Agrobacterium tumefaciens* strain GV3101. Plant transformations were conducted using the floral dip method (Clough and Bent, 1998). Transgenic plants were selected for BASTA resistance (10 mg/L) for three generations, to ensure the homozygous and non-segregating population of seedlings for each construct.

Agar-based plate experiments for mutant lines

Arabidopsis seeds were sterilized for 10 min with 50% bleach and rinsed five times using milli-Q water, and germinated on ½ strength Murashige and Skoog (MS) medium containing 0.5% (w/v) sucrose, 0.1% (w/v) 4-morpholineethanesulfonic acid (MES), and 1% (w/v) agar, with pH adjusted to 5.8 with KOH. After 24 h of vernalization at 4°C in the dark, the plates were placed in the Conviron growth chamber with the light intensity of 130–150 $\mu\text{mol} \times \text{m}^{-2} \times \text{s}^{-1}$ in a 16 h light / 8 h dark cycle at 21°C and 60% humidity. For plate assays, 4 days after germination, the seedlings were

transferred to ½ MS media supplemented with 0, 75, or 125 mM NaCl as indicated. The plates were scanned using EPSON scanner every other day, starting from 4 days after germination until the plants were 14 days old. To analyze root system architectural traits from the scanned plate images, we used SmartRoot plugin (Lobet et al., 2011 [DOI](#)) in ImageJ to trace the root, and extracted root-related features in the CSV format followed by data analysis in R (<https://rpubs.com/mri23/913282> [DOI](#)).

Evaluation of plant performance under salt stress in soil

The seeds were germinated in ½ MS media for one week, as described for the agar-based plate experiments. One week after germination, the seedlings were transplanted to the pot (12 x 4 cm insert) containing the Cornell Mix soil (per batch combine: 0.16 m³ of peat moss, 20.84 kg of vermiculite, 0.59 kg of Uni-Mix fertilizer, and 2.27 kg of lime) watered to 100% water holding capacity and placed in the walk-in growth chamber with the 16 h light / 8 h dark period, 22°C and 60% relative humidity throughout the growth period. When all of the pots dried down to the weight corresponding to 50% of their water holding capacity, they were soaked for 1 h in tap water or a 200 mM NaCl solution, resulting in an effective concentration of 100 mM NaCl based on the 50% soil water holding capacity, which corresponded to a moderate level of salt stress (Awlia et al., 2016 [DOI](#)). The control pots were soaked for the same length of time in 0 mM NaCl solution, to account for the soil saturation effect. We then allowed the pots to be drained for 2-3 h to eliminate excess moisture. The pots were placed under phenotyping rigs equipped with an automated imaging system (Yu et al., 2023) and the pot weight was measured daily to maintain the reference weight corresponding to 50% of the soil water holding capacity throughout the experiment. We would like to note that this gravimetric based method for application of salt stress has been developed for soils typically used for pot-grown plants, with relatively high water holding capacity (Awlia et al., 2016 [DOI](#)). Within these specific conditions, no drought stress symptoms were observed.

To evaluate the growth of soil-grown plants, we used an in-house developed automated imaging system using Raspberry Pi cameras to capture the growth dynamics over the period of two weeks (Yu et al., 2023). The collected images were processed using the PlantCV pipeline (Gehan et al., 2017 [DOI](#)), followed by the data analysis in R, as described here (https://rpubs.com/mjulkowska/sr3g_Maryam2021OctNov [DOI](#)).

Transient expression of SR3G constructs in tobacco epidermal leaf

Agrobacterium tumefaciens strain GV3101 containing cellular marker free GFP (35S::GFP with rifampicin and spectinomycin resistant), generously provided by Professor Maria Harrison Lab at Boyce Thompson Institute, or mVenus-SR3G constructs (including UBQ10::mVenus-SR3G-full length, UBQ10::mVenus-SR3G-fragment A, UBQ10::mVenus-SR3G-fragment B, and UBQ10::mVenus-SR3G-fragment AB, all with spectinomycin and tetracycline resistant) were grown in liquid YEP medium with appropriate antibiotics overnight at 28°C while shaking. The following concentrations were used for antibiotics when used for transient transformation: spectinomycin (50 ug/ml), tetracycline (10 ug/ml), and rifampicin (12.5 ug/ml). Rifampicin and kanamycin (50 ug/ml) were used for P19 plasmid (generously provided by Maria Harrison Lab at BTI) as a silencing inhibitor. The following day, OD₆₀₀ was measured to be 0.3-0.5. The cultures were then centrifuged at 8000x g for 2 min. The *Agrobacterium* pellet was resuspended in an induction buffer containing 10 mM MgCl₂, 10 mM MES, pH 5.6, and supplemented with 200 µM acetosyringone and was incubated for 2-4 h in the dark at room temperature. Leaves of 4-weeks-old *Nicotiana benthamiana* plants were infiltrated using a syringe. After 48 hours, the fluorescence was visualized in epidermal cells of leaf discs using confocal microscopy.

Confocal microscopy images were taken using the Leica TCS SP5 Laser Scanning Confocal Microscope with the following setting: Excitation at wavelengths of 488 nm (GFP), 514 nm (mVenus), and 561 nm (chlorophyll autofluorescence) were provided using an argon or DPSS

(Diode-Pumped Solid State) laser. A spectral emission range of 505–525 nm for GFP, 526–560 nm for mVenus, and 680–700 nm for chlorophyll autofluorescence was used.

Elemental analysis

The elemental analysis was performed using Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES). For analysis of Na⁺ and K⁺ ions in roots and shoots of plate-grown seedlings, the plants were grown on ½ x MS plates as described above. After two weeks of exposure to 0, 75 or 125 mM NaCl, root and shoot tissues were harvested and measured for fresh weight, rinsed in milli-Q water and collected into separate paper bags that were dried at 60 °C for 2–3 days. Subsequently, the dry weight was recorded. Samples were digested in double distilled HNO₃, followed by the addition of 60/40 nitric/perchloric acid continuing incubation at 150 °C. Samples were processed for ICP-AES analysis using a Thermo iCap 7000 ICP-AES after diluted to 10 ml with deionized water. For soil-grown plants, the entire rosette leaves were harvested at the end of week four and underwent a similar procedure as described above. Ion content was calculated per dry weight for each sample, followed by data analysis in R (sr3g:<https://rpubs.com/mri23/913285> , wrky75/sr3g double mutants: <https://rpubs.com/mri23/1157092>  for plate-grown and sr3g:<https://rpubs.com/mri23/913299>  for soil-grown seedlings).

Electrolyte leakage analysis

The electrolyte leakage measurement was done as previously described (Hu et al., 2015 ) with slight modifications. In brief, for each plant, three leaf discs were incubated in 2 ml of distilled water for 24 h at the light at room temperature with gentle shaking, followed by measuring the initial electrolyte leakage using a conductivity meter (Horiba LAQUAtwin EC-11 Compact Conductivity Meter, # 3999960125). The samples were then subjected to 80 °C for 2 h to release the total electrolyte, and cooled at room temperature for 24 h. Final electrolyte leakage was measured the following day. The electrolyte leakage percentage was calculated by dividing initial conductivity by final conductivity * 100 and followed by data analysis in R (<https://rpubs.com/mri23/913288> ) and <https://rpubs.com/mri23/1102053> .

RT-qPCR analysis

Total RNA was isolated from Arabidopsis seedlings root and shoot treated with or without 75 or 125 mM NaCl for one week using the RNeasy Plant Mini Kit (Qiagen). RNA samples were cleaned at least twice using RNase-free DNase (Qiagen) to remove genomic DNA contamination. One microgram of total RNA was used for a reverse transcriptase reaction using iScript cDNA Synthesis Kit (Catalog #170-8891; Bio-Rad). RT-qPCR analysis was conducted using iQ SYBRGreen Supermix (Bio-Rad) according to manufacturer's instructions in the CFX96 real-time PCR system (Bio-Rad). *AT4G04120* (*transposable_element_gene*) was used as a housekeeping gene for data normalization. RT-qPCR experiments were conducted using three independent experiments, each with at least three biological replicates. The list of primers used is shown in **Table S4**.

Suberin staining

Suberin content was examined in 6-days-old Arabidopsis roots treated with or without 75 mM NaCl for two days using Fluorol Yellow 088 (FY; Santa Cruz Biotechnology Incorporation) staining. Seedlings grown on agar plates (as described above for mutant screens) were incubated in a freshly prepared solution of FY (0.01% [w/v] in lactic acid 85% [W/W]) at 70 °C for 30 min, rinsed in milli-Q water for three times with 5 min each, and mounted on slides with 50% glycerol before confocal microscopy examination using excitation at wavelength of 488 nm and a spectral emission range of 500–550 nm. Suberin quantification was performed using the ImageJ software followed by data analysis in R (<https://rpubs.com/mri23/935019> .

Suberin monomer quantification using Gas Chromatography–Mass Spectrometry (GC-MS)

100 – 150 *Arabidopsis* seedlings were grown on the plates for four complete days, transferred to 75 mM salt plate for two days, and roots were harvested in distilled water to remove any remaining agar. Each plate was treated as one biological replicate, and five plates were used for each genotype/condition. Each plate of roots was collected into a microcentrifuge tube and dried in a speed vacuum concentrator (Thermo Electron, Waltham, MA), and weighed. Approximately 1 mg of dried root were collected from each plate. Suberin monomer quantification was performed according to (Delude et al., 2017 [DOI](#); Kreszies et al., 2019 [DOI](#)) with modifications. Dried root tissue was incubated in 1% cellulase plus 1% pectinase in 10 mM citrate buffer (pH 3.0) for two weeks, with the incubation solution changed every three days. After the cellulase/pectinase solution was removed, chloroform/methanol solution (2:1 and 1:2, v/v) were mixed with the root tissue for three days respectively, to remove lipids. The remaining root tissue was mixed with 1 ml freshly prepared 5% H₂SO₄ in methanol (v/v) and 200 µl of toluene containing 20 µg of heptadecanoic acid as internal standard, sealed in an 8 ml capped tube and heated at 85°C for 90 min. After samples were cooled to room temperature, 200 µl hexane and 1.5 ml 0.9% sodium chloride in water are added to the capped tube with shaking. After centrifuging at 1000 *g* for 1 min, the upper organic phase was transferred to a glass insert of GC vial and dried. 40 µl pyridine was added to the glass insert, 40 µl *N,O*-bis(trimethylsilyl)trifluoroacetamide (Restek, Bellefonte, PA) was added subsequently, incubated at 65°C for 2 h. Standards of 1 µl derivatized sample was injected into GCMS (Agilent 6890-5975, Santa Clara, CA), with the inlet temperature of 250°C, column helium flow of 1 ml/min, and a split ratio of 5:1. The oven temperature was held at 50°C for 1 min, raised to 200°C at the rate of 25°C/min, held at 200°C for 1 min, raised to 320°C at the rate of 10°C/min, and finally held at 320°C for 5 min. 1-Docosanol (22:0-OH), 1-eicosanol (20:0-OH), hexadecanedioic acid (16:0-DCA), 1-hexadecanol (16:0-OH), 16-hydroxy hexadecanoic acid (16:0-ωOH), C4 – C24 Even Carbon Saturated FAMES (Sigma 49453-U), 20-hydroxy arachidic acid (20:0-ωOH) (Cayman Chemical Company, Ann Arbor, Michigan) standards were input at 2.5 µg, 5 µg, 10 µg, and 20 µg into a mixture and processed with the same procedure as the tested samples to build the standard curve. The MS spectrum of derivatized products with ±2n carbon number of these standards was calculated and identified, referring to the retention order in (Delude et al., 2017 [DOI](#)).

Yeast One-hybrid Assay

The respective combinations of *pB42AD*-fusion effectors and *pLacZ* reporters were co-transformed to yeast strain EGY48 (Shanghai Maokang, China). Transformants were selected and grown on SD/-Trp-Ura medium. The putative interactions were observed on SD/-Trp-Ura + X-Gal after a 3d-incubation at 30°C. Yeast transformation and growth assay were performed as described in the Yeast Protocols Handbook (BD Clontech, USA). In brief, 631 bp of the *DUF247-150* (AT3G50150) and 953 bp of the *SR3G* (At3g50160) promoter region were separately ligated into *pLacZ* vector as the reporter, while the CDS of *WRKY75* (AT5G13080) was cloned into *pB42AD* vector as the effector. The *SR3G::pLacZ* and *WRKY75-pB42AD* were transformed into the yeast strain EGY48 (Shanghai Maokang, China). The respective combinations of *pB42AD*-fusion effectors and *pLacZ* reporters were co-transformed into yeast strain EGY48. Transformants were selected and grown on SD/-Trp-Ura medium. The putative interactions were observed on SD/-Trp-Ura + X-Gal after a 3d-incubation at 30°C. Yeast transformation and liquid assay were conducted as described in the Yeast Protocols Handbook (BD Clontech, USA). The primers used for the Y1H assay are shown in **Table S4**.

Results

Natural variation in salt stress-induced changes in root:shoot ratio

To investigate salt stress-induced changes in root:shoot ratio, we developed a custom tool that quantifies green and white pixels, corresponding to shoot and root of *Arabidopsis* seedlings grown on agar plates (**Fig. 1**). The projected area of root and shoot showed high correlation with root and shoot fresh weight (**Fig. S1**), indicating that projected root and shoot area can be used as an accurate proxy to non-destructively measure the increase in root and shoot biomass from the images of seedlings grown on agar plates.

Using our custom tool, we re-analyzed the data consisting of 360 *Arabidopsis* accessions grown on plates supplemented with 0, 75, and 125 mM NaCl (Julkowska et al., 2017), for salt stress-induced changes in the root:shoot ratio (**Fig. 1**, **Fig. S2**). The growth of root and shoot for each plant was estimated using the exponential function. Salt stress-induced reduction in growth rate was observed to be dose-dependent for both root and shoot (**Fig. 1A-B**). In general, the root growth was more sensitive to salt stress compared to shoot growth, with the population-wide relative growth rates reduced to a fraction of 0.67 and 0.41 of the growth rates observed at control for 75 and 125 mM NaCl, respectively. In comparison, the growth rates of the shoot were significantly reduced to 0.71 and 0.43 of the control in 75 and 125 mM NaCl treatments, respectively (**Fig. S3**). While the mean value of root:shoot growth rate did not change upon salt stress treatment, the variance in the root:shoot ratio significantly expanded with the increasing concentrations of salt (**Fig. 1C**). These results suggest that while root and shoot growth are well coordinated under non-stress conditions, salt stress exposure results in loss of coordination of organ growth across *Arabidopsis* accessions. To test whether there is a clear directional correlation between the change in root:shoot ratio and overall salt stress tolerance, we have used the overall seedling size as a proxy for plant salt tolerance (**Fig. S4, S5**). No significant correlation was found between the root:shoot growth ratio and total seedling size (**Fig. S4, S5**), indicating that the relationship between coordination of root and shoot growth and salt tolerance during the early seedling establishment is complex.

Identification of genetic components underlying salt-induced changes in root:shoot ratio

To identify genetic components underlying salt-induced changes in root:shoot ratio, we used the collected data as an input for GWAS. The associations were evaluated based on the p-value, the number of SNPs within the locus, and the number of traits associated with individual loci. As Bonferroni threshold differs depending on the minor allele count (MAC) considered, we identified significant associations based on a Bonferroni threshold for each subpopulation of SNPs based on MAC (**Table S3**). While we conducted a GWAS on directly measured traits, as well as their Salt Tolerance Index (STI) values, however the amount of associations with STI was much lower compared to directly measured traits (**Table S3**). This observation aligns with the understanding that plastic responses to environmental conditions tend to be genetically more complex. This complexity likely stems from the involvement of more genetic regulators compared to low-plasticity phenotypes.

The initial GWAS was performed using 250 k SNPs (Horton et al., 2012), and led to the identification of 14 significantly associated loci, of which 3 and 7 loci were associated with traits measured at 75 or 125 mM NaCl treatments, respectively. We identified one salt specific association on chromosome 2 (**Fig. S6**), where two SNPs were identified within AT2G27300 encoding NAC-domain containing transcription factor (NTL8). Additionally, salt-induced changes in root:shoot ratio and shoot:total seedling area were associated with SNPs on chromosome 3, where 2 significantly associated SNPs were flanking AT3G22430, encoding an unknown gene encoding a domain of unknown function (**Fig. S7**).

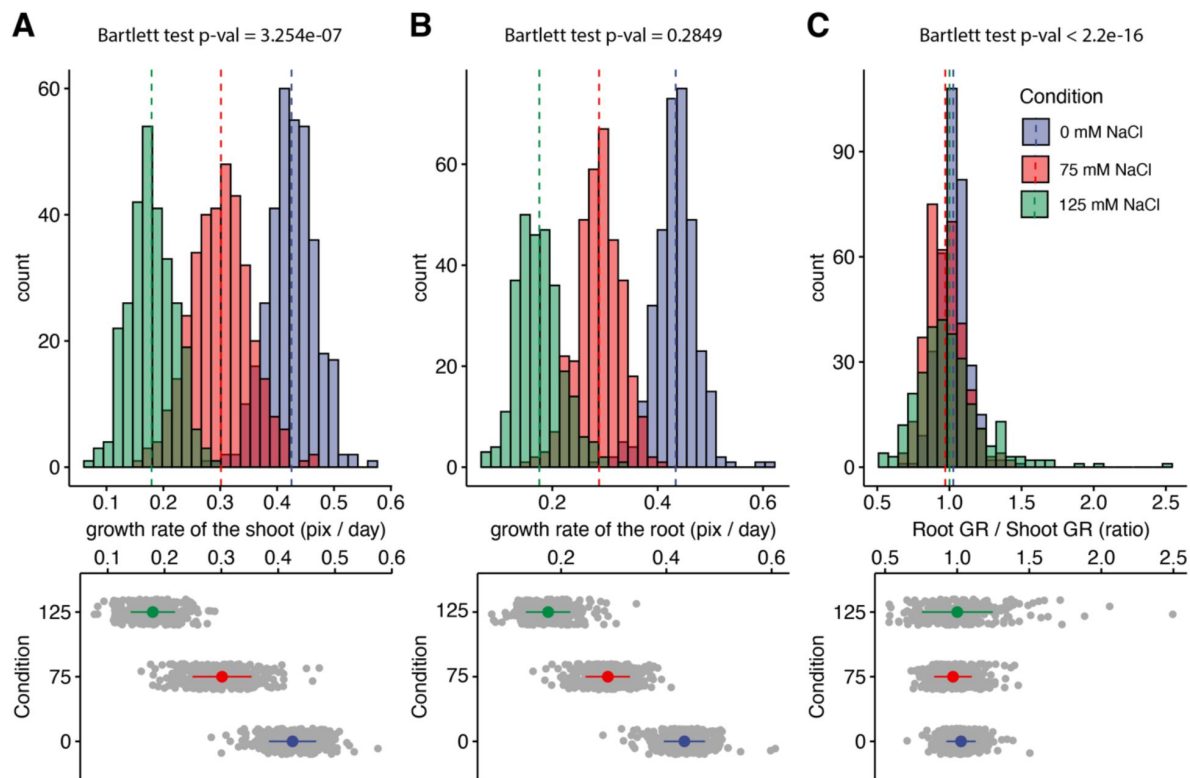


Figure 1.

Salt stress is causing discoordination of root and shoot growth.

Arabidopsis HapMap population was screened for salt stress-induced changes in root:shoot ratio. The increase in the projected area of shoot and root (Fig. S2) was used to estimate **(A)** shoot and **(B)** root exponential growth rate. **(C)** The root:shoot growth rate ratio was calculated per genotype. The histograms represent the number of accessions across three studied conditions (0, 75 and 125 mM NaCl), whereas the population average is indicated using the dashed line. Additionally, the distribution of the genotypes within each treatment was visualized using the error plots (lower panel), where population average and standard error is indicated using a colored point and a line, respectively. Individual grey points represent individual genotypes.

To refine our analysis and further prioritize candidate genes for validation, we repeated the GWAS using a higher-density 4M SNP panel (1001 Genomes Consortium, 2016 [\[16\]](#)). This increased SNP density provided more associations and highlighted loci with greater confidence due to the higher resolution. The higher-density panel also allowed us to capture different linkage disequilibrium (LD) blocks across the genome. One locus on chromosome 2, detected by both the 250K and 4M SNP panels, appeared to be in a region of strong LD or possibly under selection, consistently identified by both panels. However, most loci were not detected by the lower-density 250K panel, likely due to differences in the panels' resolution and statistical power.

The new GWAS with the 4M SNP panel revealed 32 additional loci, with only one locus (locus 30, **Table S3**), being detected by both the 250K and 4M SNP panels. We prioritized the identified loci based on several factors, including the number of significantly associated SNPs, MAC, and the number of traits associated with each locus. This led to the identification of three loci of major interest (**Table S4**). We further examined the regions in LD with the significantly associated SNPs for sequence divergence and open reading frames (**Fig. 2C**, **Fig. S8**, **Fig. S9**).

Out of identified associations, locus 30 stood out as it consisted of nine SNPs associated with p-value above the Bonferroni threshold, and was associated with both root:shoot and shoot:total seedling area at 6 and 8 days after transfer of the seedling to 125 mM NaCl (**Fig. 2A-B**, **Fig. S8**, **Table S4**). The significantly associated SNPs were located in and around the gene coding region of AT3G50160, which encodes a domain of unknown function 247 (DUF247). We investigated the region for sequence divergence, and found that the region contained not only a high degree of sequence diversity but also potential structural variants (**Fig. 2C**). To identify allelic variation underlying the initial associations, we performed haplotype analysis, where accessions with shared SNP pattern within the protein-coding part of AT3G50160 were grouped into individual haplotype-groups (**Fig. 2D**). Each unique haplotype, represented by at least three accessions, was subsequently studied for differences in observed phenotypes. We observed that Haplotype 1, represented by 46 accessions including Col-0, was significantly different from Haplotype 11, represented by 11 accessions including Blh-1, in shoot-per-total seedling area at 6 days after transfer to 125 mM NaCl (**Fig. 2D**), as well as shoot growth rate and relative shoot growth rate at 125 mM NaCl (**Fig. S9**). Upon further investigation and sequencing of AT3G50160 in both Col-0 and Blh-1 accessions, we discovered that the Blh-1 allele carries a large insertion at 644 nucleotides downstream from the ATG of AT3G50160, followed by multiple other sequence variations (**Fig. S10**). The first insertion leads to a premature STOP-codon within the DUF247 domain after Gly-215 (**Fig. 2E**, **Fig. S11**), resulting in a truncated protein that still carries a nuclear localization signal and coiled-coil domain, but is missing a major part of the calcium-binding domain and a predicted transmembrane domain (**Fig. S12**). As there are other DUF247s within the locus associated with salt-induced changes in root:shoot ratio, we decided to name the gene containing majority of the associated SNPs (At3g50160) as Salt Root:shoot Ratio Regulator Gene (SR3G). Together, these results suggest that the allelic variation in SR3G protein sequence might underlie the observed differences in root:shoot ratio under salt stress conditions.

Evolutionary context of SR3G locus reveals its origin in local genome duplication

SR3G sits in what appears to be a tandemly duplicated array of DUF247s along Arabidopsis chromosome 3. The genome-wide identification of SR3G included eight DUF247 paralogs, AT3G50120, AT3G50130, AT3G50140, AT3G50150, AT3G50170, AT3G50180, AT3G50190, and AT3G50200. Reciprocal best BLAST within CoGe's BLAST tool of these nine DUF247 paralogs against the genomes of seven other Brassicaceae (see **Table S5**) revealed a total of 50 potential orthologs used for downstream analysis. While the copy number of DUF247 loci are similar across the examined species, the noted exceptions are those with recent whole genome duplication events

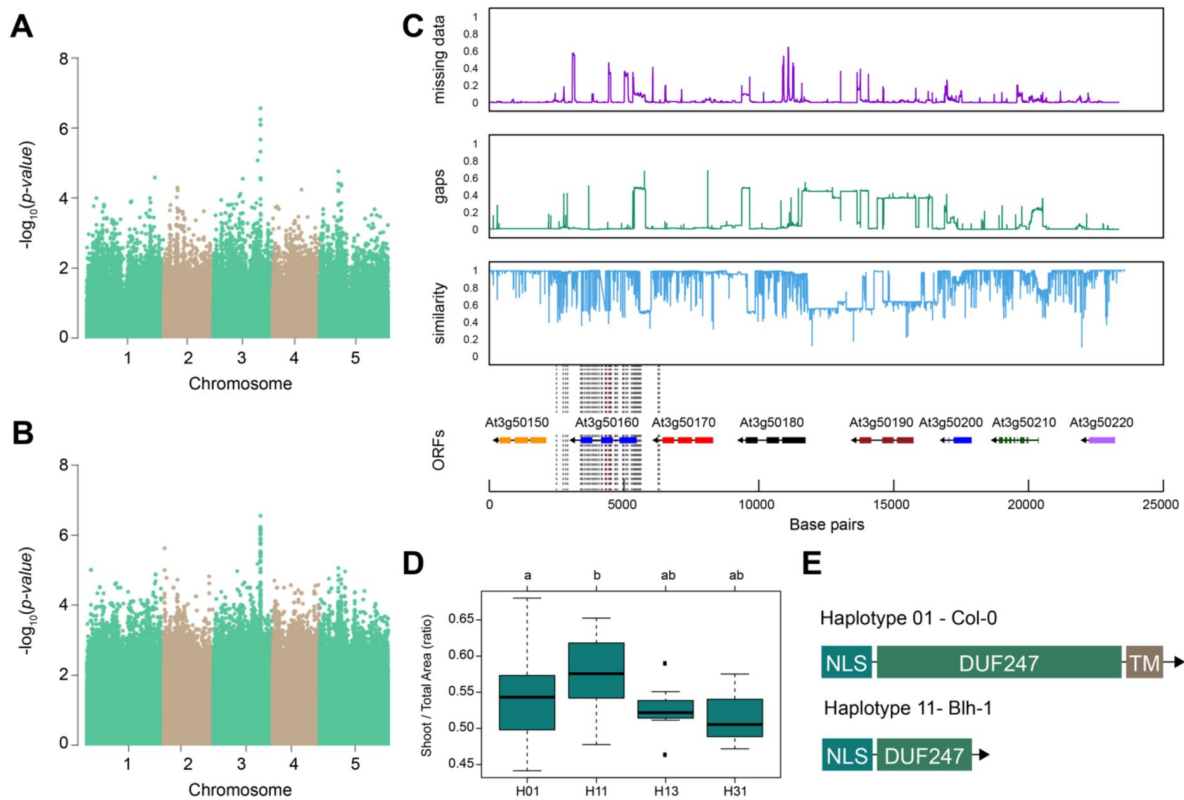


Figure 2.

SR3G is associated with salt stress-induced changes in shoot-per-total seedling area ratio.

The salt induced changes in root-to-shoot and shoot-per-total seedling area ratios were used as an input for GWAS. Manhattan plots representing the associations of shoot-per-total seedling area recorded 6 days after transfer to 125 mM NaCl and **(A)** 250 k SNPs and **(B)** 4 M SNPs. **(C)** Significant associations were found with the SNPs forming a locus on chromosome 3 in and around *SR3G* (AT3G50160), encoding Domain of Unknown Function 247 (DUF247). The natural variation in the LD region was studied in 162 accessions sequenced by 1001 Genomes Project. The upper panel represents portion of the missing data, upper middle panel represents deletions relative to Col-0, while lower middle panel represents the sequence similarity compared to Col-0. The bottom panel represents the open reading frames (ORFs) within the LD, and the location of associated SNPs is indicated with the dashed lines. Red dashed lines represent associations above Bonferroni threshold in 250k SNP mapping, while grey lines represent associations with $-\log_{10}(p\text{-value}) > 5$ in the 4M SNP mapping. **(D)** The haplotype analysis performed using SNPs located within the coding region of the *SR3G* revealed significant differences in shoot-per-total seedling area recorded 6 days after transfer to 125 mM NaCl between Haplotype groups 1 (represented by 46 accessions, including Col-0) and 11 (represented by 11 accessions, including Blh-1). The significant differences between individual haplotype groups were tested using ANOVA with Tukey HSD to identify significantly different groups. **(E)** Upon further sequencing of *SR3G* in accessions from Haplotype 1 and 11, revealed two 200 bp insertions within *SR3G* exons in 4 tested accessions belonging to haplotype 11 group. These 200 bp insertions resulted in a premature STOP-codon within the DUF247 domain (after Gly-215).

(*Brassica rapa* and *Camelina sativa*). Given these data, it appears that DUF247 loci are commonly tandemly duplicated, although the eight paralogs observed in *Arabidopsis* represents an extreme case for this gene family.

To examine conservation within the predicted DUF247 functional domains, we generated an amino-acid based multiple sequence alignment using MUSCLE. All of the identified DUF247 proteins contained the DUF247 domain near the center of the translated sequence. While partial deletions were identified outside of the DUF247 domain at the C-terminus of a few DUF247s, such as *CARHR159180*, *Csa09g052970*, and *Bra01902*, in general the DUF247 domains were highly conserved across Brassicaceae. The amino acid sequences spanning the “coiled-coil domain” (Amino acid position: Met153-Lys173), putative Ca^{2+} binding domain (Amino acid position: Gly210-Phe372), and super polar fragment (Amino acid position: Ser301-Ile320) were further selected for phylogenetic reconstruction (**Fig. 3B**). The resulting phylogeny demonstrated that SR3G is sister to a clade containing AT3G50150 and representatives from *C. sativa*, *Capsella rubella*, and *A. lyrata*, with representative DUF247s from *B. rapa* and *Eutrema salsugineum* serving as a monophyletic outgroup. These data support a hypothesis where SR3G is a recent duplicate (likely A/C Clade specific; Forsythe et al., 2020) of AT3G50150 and that representative SR3G sequences in *C. sativa*, *C. rubella*, and *A. lyrata* have been lost.

Given the specific retention of AT3G50150 in *A. thaliana*, we next tested whether this locus is experiencing directional selection. To test this assumption, we grouped the branches leading to SR3G into four groups (**Figure 3B**: ω_0 , ω_1 , ω_2 , and ω_{rest}) and then performed a branch-site model A hypothesis test. In our null hypothesis (H_0), we hypothesized that all branches have the same selection pressure (H_0 : $\omega_{\text{rest}} = \omega_0 = \omega_1 = \omega_2$; see **Table S6** for omega values for each hypothesis). For our alternative hypothesis, we created five different ω groups to cover scenarios where specific branches may be experiencing directional selection (See Methods). Based on our analysis, the branch leading to SR3G is experiencing directional selection (likelihood score = 20625.5, Chi-square derived P value; $P = 0.00418$). Interestingly, no other branches appeared to be experiencing the same selection. To pinpoint specific amino acids within SR3G under selection, we performed the Bayes Empirical Bayes (BEB) test, identifying three sites with a posterior probability > 0.90, including 124C ($P = 0.992$), 206A ($P = 0.951$), and 209A ($P = 0.988$). Two out of three sites are located at the putative Ca^{2+} binding site and all three sites possess a unique amino acid compared with the other 49 amino acid sequences (**Fig. 3C**).

SR3G expression is enhanced in root by salt stress

To identify the developmental window and specific plant tissues where the *SR3G* gene is active, we examined its expression pattern. Based on *Arabidopsis* eFP Browser (<http://bar.utoronto.ca/>), *SR3G* expression is relatively low, and restricted to the early stages of seedling development (**Fig. S13A**). Upon examining the expression profiles specific to different tissues, we observed that *SR3G* is predominantly expressed in the root stele (**Fig. S13B**). Moreover, *SR3G* expression was induced by salt stress (**Fig. S13C**). To validate these results, we performed RT-qPCR on 11 days old seedlings of Col-0 exposed to 0 or 75 mM NaCl for 7 days. In Col-0 seedlings, the transcript levels of *SR3G* were significantly higher in shoot compared to the root tissue (**Fig. 4A**). Interestingly, the *SR3G* transcripts increased only in root upon salt stress exposure but not in shoot (**Fig. 4A**). These results suggest that *SR3G* is the most prominent DUF247 studied within our study to affect root development under salt stress.

We further examined expression profiles of other DUF247 genes that directly neighbor *SR3G* in response to salt stress using the eFP Browser. We observed similar expression patterns as *SR3G* for other DUF247s including At3g50150, indicating their responsiveness to salt stress (**Fig. S14**). Because DUF247-150 (At3g50150) is closely located to the *SR3G* in chromosome 3, we investigated its gene expression in young seedlings. We observed that its expression was significantly higher in shoot compared to the root tissue. However, its expression remained unchanged in both root and shoot under salt stress (**Fig. 4B**). For the remaining DUF247 genes (At3g50120 and At3g50200),

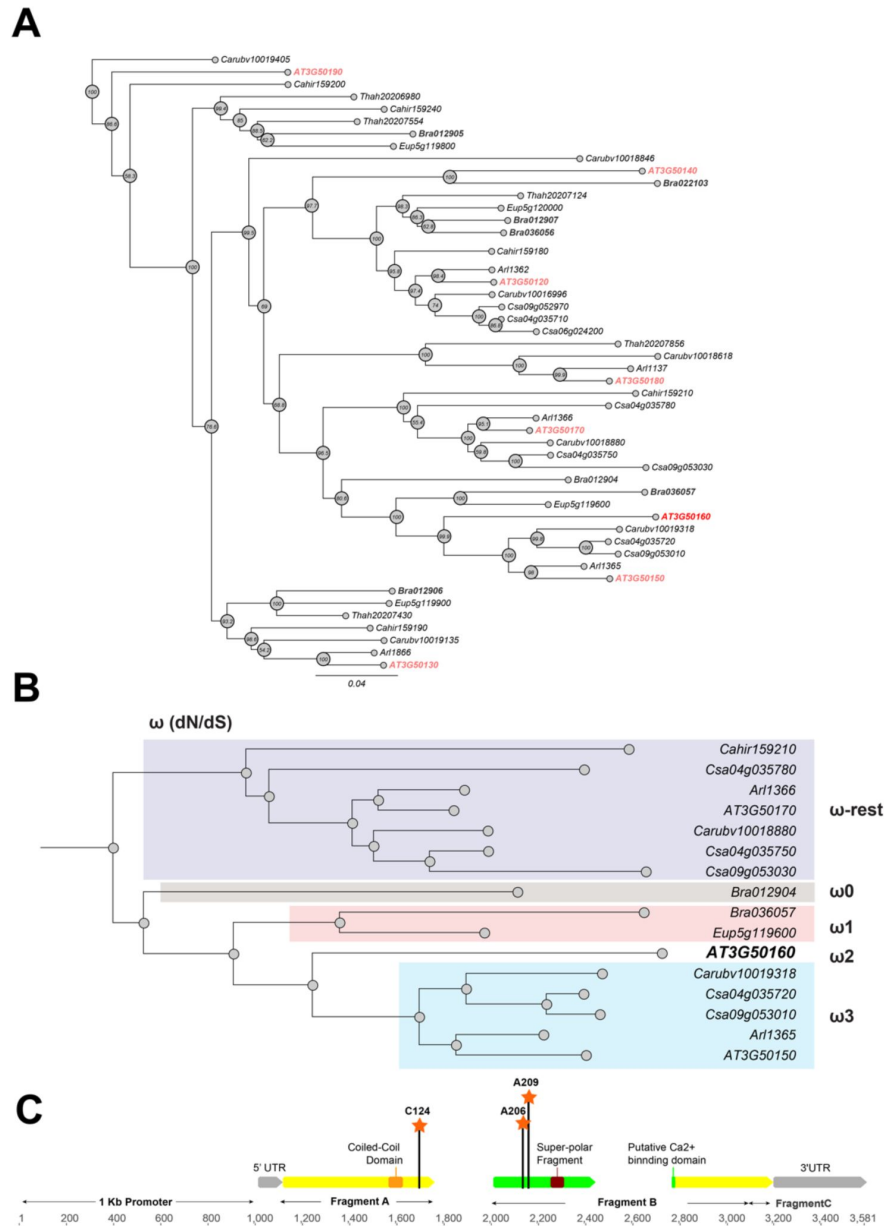


Figure 3.

Phylogeny and positive selection analysis of *SR3G* and its orthologs from other species.

The orthologous genes of *SR3G* (AT3G50160) from other seven species were identified using GoGeBlast (E-value < 1.00E-10).

(A) The protein sequences of the 50 homologous genes were aligned by MUSCLE and then an unrooted phylogenetic tree was reconstructed by a Maximum -likelihood (ML)-derived style by RAxML (Bootstrap number: 100). **(B)** A positive selection analysis was conducted by branch-site model A test: Specifically, we hypothesized the differential or constant dN/dS (ω) substitution rate among the closely distant branches leading to DUF247 (AT3G50160) (See Method & Materials for details). These branches were marked as ω_1 (light brown branch), ω_2 (DUF247-specific branch), ω_3 (light blue branch), and ω -rest (light purple branch). **(C)** The gene structure of DUF247 (AT3G50160) was illustrated by wide arrows (grey arrow: 5 and 3' UTR, yellow arrow: coding-region). The regions used for cloning (Fragment A, B, and C) were marked by narrow arrows along with "Coil-Coil Domain" and "Super-polar Fragment" marked on Fragment A and B, respectively. Lastly, the Bayes Empirical Bayes (BEB) was performed to test the probability of sites along with $\omega > 1$ over the DUF247 (AT3G50160). Three sites were generated with a posterior probability > 0.90: 124C (P = 0.992), 206A (P = 0.951), and 209A (P = 0.988). These three sites were marked with an orange asterisk and assumed to have been undergoing positive selection.

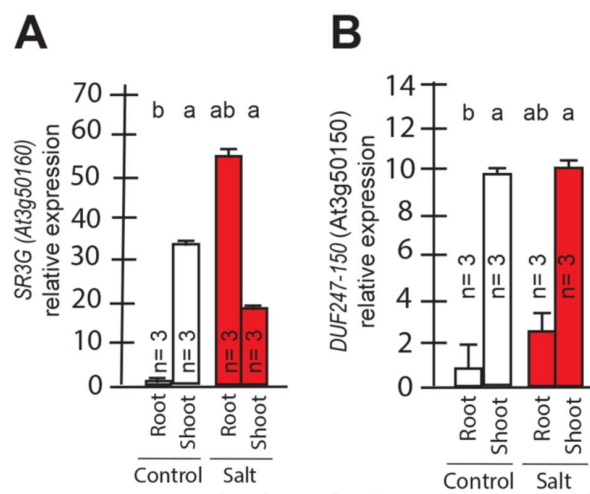


Figure 4.

Transcript abundances for the two closely related *DUF247s*.

Expression of **(A)** *SR3G* (AT3G50160) **(B)** and its closest homologue *DUF247-150* (AT3G50150) were measured in Col-0 seedlings grown with and without salt stress. RT-qPCR analyses were conducted using seedlings grown on 1/2x MS for 4 d and then followed by transferring to the treatment plates with or without 75 mM NaCl for one week. Mean values are shown $\pm$ SE, with number of replicates (n) shown in each graph. *AT4G04120* was used as a reference gene for normalization. Significance was determined by the Tukey-Kramer HSD test in JMP.

we observed no significant changes, while others (At3g50130, At3g50140, At3g50170, At3g50180) showed reduced expression upon exposure to salt stress (Fig. S14). These results suggest that *SR3G* is the most prominent DUF247 studied herein involve in regulation of root development under salt stress.

SR3G relocates from plasma membrane to nucleus/cytosol after removal of its transmembrane domain

SR3G protein is predicted to contain a nuclear localization signal (Lys15-Ala19), a coiled-coil domain (Met153-Lys173), a Ca^{2+} binding pocket (Gly210-Phe372), and a trans-membrane domain (Pro469-Thr503) (Fig. S15). We generated constitutive expression constructs of full-length *SR3G* gene CDS, and a combination of individual domains (refer to as A, B, and AB domains) to identify their subcellular localization (Fig. S15). Fragment A comprised a nuclear localization signal, a coiled-coil domain, and the initial segment of the Ca^{2+} binding pocket, encompassing seven amino acids. Fragment B encompassed the remaining segment of the Ca^{2+} binding pocket. Fragment C only included the transmembrane domain. All constructs contained N-terminal fusion of mVenus. The individual constructs were used in transient expression assay in tobacco leaves. Our result showed that full-length *SR3G* localizes to the plasma membrane (Fig. 5A) as its signal overlapped with FM4-64, the plasma membrane marker (Fig. 5B). The *SR3G* fragments from which the transmembrane domain was excluded—namely fragments A, B, and AB—exhibited localization in both the nucleus and cytosol, resembling the pattern observed with free-GFP localization (Fig. 5C-F). It is noteworthy that while the signal from fragment A was directed to both the cytosol and nucleus, its intensity was notably stronger in the nucleus (Fig. 5C), aligning with the prediction that it contains the nuclear localization signal. Together, these findings indicate that the positioning of *SR3G* within the plasma membrane is due to the transmembrane domain encoded within the fragment C, and its removal could potentially release the protein into both the nucleus and cytosol.

Lower expression of *SR3G* results in reduced lateral root development under salt stress

To investigate role of *SR3G* in salt-induced changes in root growth and development, we studied the root system architecture of two T-DNA insertion lines: *sr3g-4* (SAIL_690_E12) and *sr3g-5* (SAIL_608_C06) localized in second exon and first intron region of the *SR3G* gene, respectively (Fig. S16A). Among these two mutant lines, *SR3G* expression was reduced in the *sr3g-5* mutant compared to Col-0 (Fig. S16B), indicating the *sr3g-5* is a knockdown mutant. We did not observe any differences in *SR3G* transcript abundance between *sr3g-4* and Col-0 (Fig. S16B). Moreover, we examined the transcript abundance of the closely related gene (i.e., *DUF247-150*, At3g50150) in the *sr3g-5* mutant, and found no significant difference. In the case that *DUF247-150* would act redundantly to *SR3G*, we would expect that the expression of *DUF247-150* would be increased in *SR3G* knockdown mutant. As this is not the case, these results further contribute to the evidence that *SR3G* and *DUF247-150* are not acting redundantly (Fig. S16C).

Two T-DNA insertion lines were subsequently studied for salt-induced alterations in root architecture (Fig. 6 and Fig. S17 and S18). Among the various features of the root system architecture that we investigated (Fig. S17), *sr3g-5* exhibited a significantly reduced growth rate of lateral roots under 75 mM NaCl, but not under 125 mM NaCl, when compared to Col-0 (Fig. 6A). Likewise, the *sr3g-5* mutant displayed a significant reduction in the Stress Tolerance Index (STI), calculated as the ratio of growth rate under salt to growth rate under control conditions, for lateral root length under 75 mM NaCl (Fig. 6B). No significant difference was observed between *sr3g-4* and Col-0 (Fig. S18), except for Stress Tolerance Index (STI) calculated using growth rates of lateral root length and number, consistent with the observations made for the *sr3g-5* mutant line (Fig.

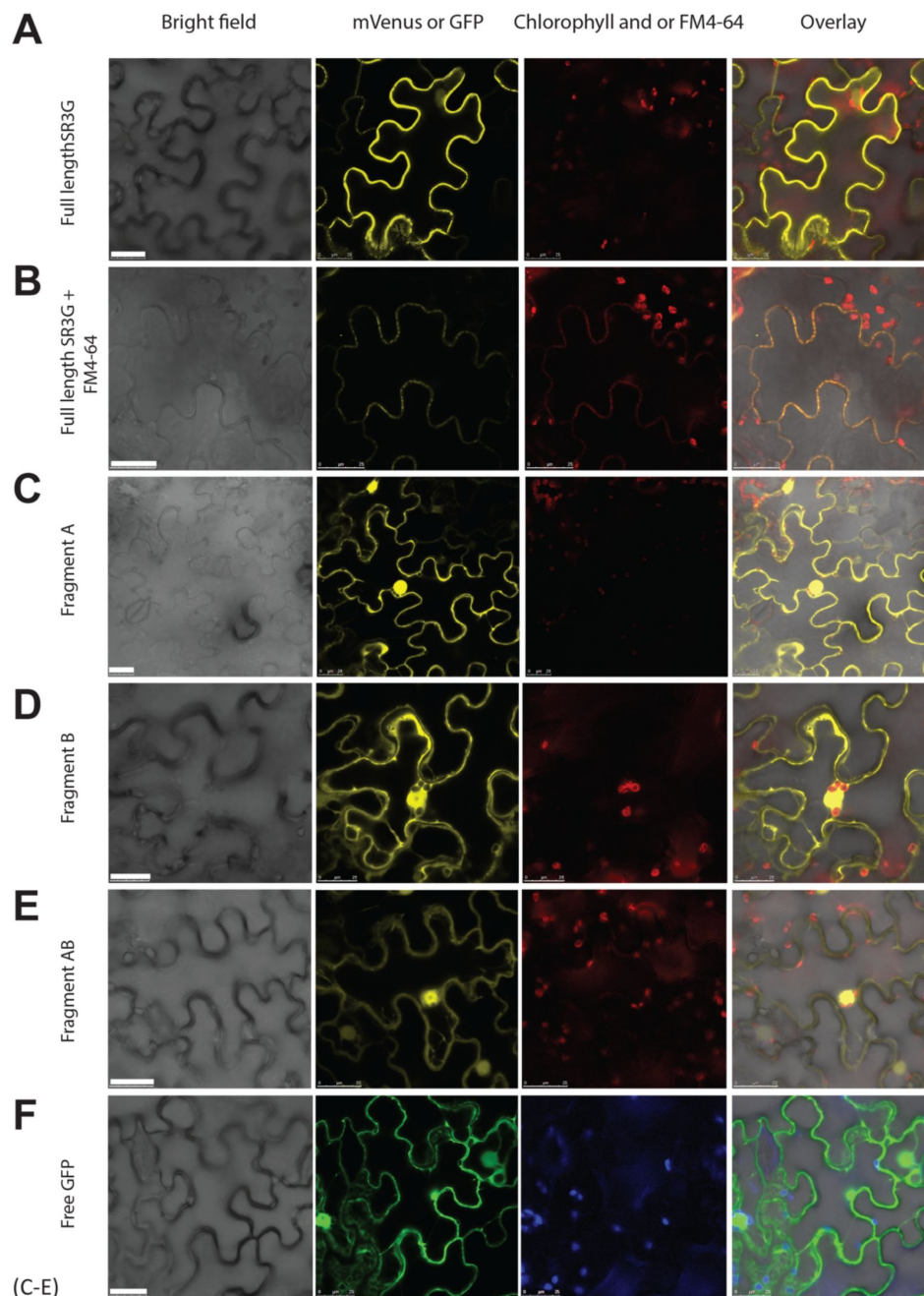


Figure 5.

SR3G full-length protein resides in the plasma membrane while removal of its transmembrane domain results in protein relocation to nucleus and cytosol.

SR3G full-length protein (**A-B**) or truncated versions (**C-E**) fused to mVenus at the N-terminus were agro-infiltrated into the *Nicotiana benthamiana* leaves for transient expression. Shown are a bright-field image of transfected leaf cells and mVenus-SR3G-mediated fluorescence as well as GFP-based subcellular marker. (**A**) Localization of full-length SR3G alone or (**B**) in combination with the known plasma membrane dye, FM4-64. Localization of truncated versions of SR3G are shown for (**C**) SR3G-Fragment A, (**D**) SR3G-Fragment B, and (**E**) SR3G-Fragment AB. (**F**) Free GFP was used as a nuclear and cytosolic marker. Scale bar = 25 μ m.

6B [↗](#)). Based on the RT-qPCR results (**Fig. S16**), we proceeded to further investigate the role of *SR3G* in salt stress, focusing on the *sr3g-5* mutant in our subsequent investigations. As Na^+ exclusion from the shoot is one of the main mechanisms of salt tolerance (Julkowska et al.,

2017; Møller et al., 2009 [↗](#); Munns and Tester, 2008 [↗](#)), we further investigated the role of *SR3G* in ion exclusion using plate-grown seedlings. We used root and shoot samples collected for Col-0 and *sr3g-5* mutant to examine Na^+ and K^+ accumulation using ICP-AES at 14 days after exposure to salt stress. We observed no significant differences between Col-0 and *sr3g-5* mutant for Na^+ and K^+ accumulations and their corresponding ratio upon salt stress exposure (**Fig. S17**, **Fig. 6C** [↗](#)). These results suggest that *SR3G* plays a role in root development under the salt, but does not affect ion accumulation at early developmental stages.

To evaluate the effects of other closely related DUF247 genes on root development, we have investigated the T-DNA insertion lines targeting the neighboring DUF247 genes (AT3G50120, AT3G50130, AT3G50140, AT3G50150, AT3G50170, AT3G50180, AT3G50190, AT3G50200), and RNAi lines targeting one or more DUF247s within the interval (**Table S4**, **Fig. S19-S23**). For the screened T-DNA insertion lines (**Fig. S19-20**), we have not observed consistent changes in main or lateral root development in either control or salt stress conditions across alleles targeting the same gene. The only exception was *DUF247-200* encoded by AT3G50200, where two out of three studied T-DNA insertion alleles exhibited increased salt tolerance index for main root growth rate (**Fig. S20**).

For the screened RNAi lines (**Table S4**, **Fig. S21-S23**), we observed the most interesting root architecture phenotypes in lines targeting the neighboring genes of the *SR3G* (RNAi line #04, targeting AT3G50150, RNAi line #16 targeting AT3G50130, *SR3G*, and AT3G50170, and RNAi line #19 targeting AT3G50130, AT3G50140, AT3G50150, *SR3G*, AT3G50170, AT3G50180, and AT3G50190). Three out of five screened independent transformation lines from RNAi line #04 showed a decreased relative main root length (Salt / Control, CSHL #4 B2a, D4a, and G1d), whereas two lines showed a decrease in relative lateral root number (CSHL #4 D4a and F1b), and one line showed decreased lateral root length at 75 mM NaCl (CSHL #4 D4a **Fig. S21-23**). Two out of two independent transformants for RNAi line #16 showed a decreased relative main root length and increased lateral root number at 75 mM NaCl (CSHL #16 A2c and B1a, **Fig. S21-23**). In three of the four screened independent transformants of line #19 we observed an increased lateral root length at 75 mM NaCl (CSHL #19 F1a, I3c, and L1c **Fig. S21-23**). We did not observe strong reduction in lateral root development across RNAi lines targeting two closely related DUF247 genes (i.e., RNAi line #22, targeting AT3G50150 and *SR3G*), as in the single mutant (**Fig. 6** [↗](#), **Fig. S21**). The above results further support that these two genes do not act redundantly in the context of lateral root development.

Loss of *SR3G* results in increased salt tolerance in soil

To investigate *SR3G*'s contribution to salt stress tolerance beyond seedling stage and agar plate conditions, we evaluated the performance of Col-0 and *sr3g-5* mutant in soil. We exposed the soil-grown plants to salt stress and imaged the rosette size every 30 minutes for two weeks using the automated imaging platform (Yu et al., 2023). As salt-induced changes depend on the developmental stage at which the stress is applied (Julkowska et al., 2017 [↗](#)), we examined the role of *SR3G* exposed to salt stress at 2 or 3 weeks after germination, corresponding to early and late vegetative stages. We calculated a daily growth rate for each plant, using the projected rosette area. No significant difference was observed between Col-0 and *sr3g-5* under control conditions (**Fig. 7A-B** [↗](#)), indicating that *SR3G* does not play a role in plant development under non-stress conditions. Significant difference was observed between Col-0 and *sr3g-5* for plants exposed to early salt treatment (**Fig. 7A-B** [↗](#)), where *sr3g-5* mutant displayed an increased rosette growth rate as compared to Col-0 (**Fig. 7A** [↗](#)). Late stress exposure resulted in significantly higher *sr3g-5* growth rate at two and three days after exposure to salt stress (**Fig. 6A** [↗](#)). Similarly, the *sr3g-5* mutant showed a significantly higher rosette area compared to Col-0 exclusively when plants were exposed to salt stress at an earlier stage of their development (**Fig. 7A** [↗](#)). No significant difference

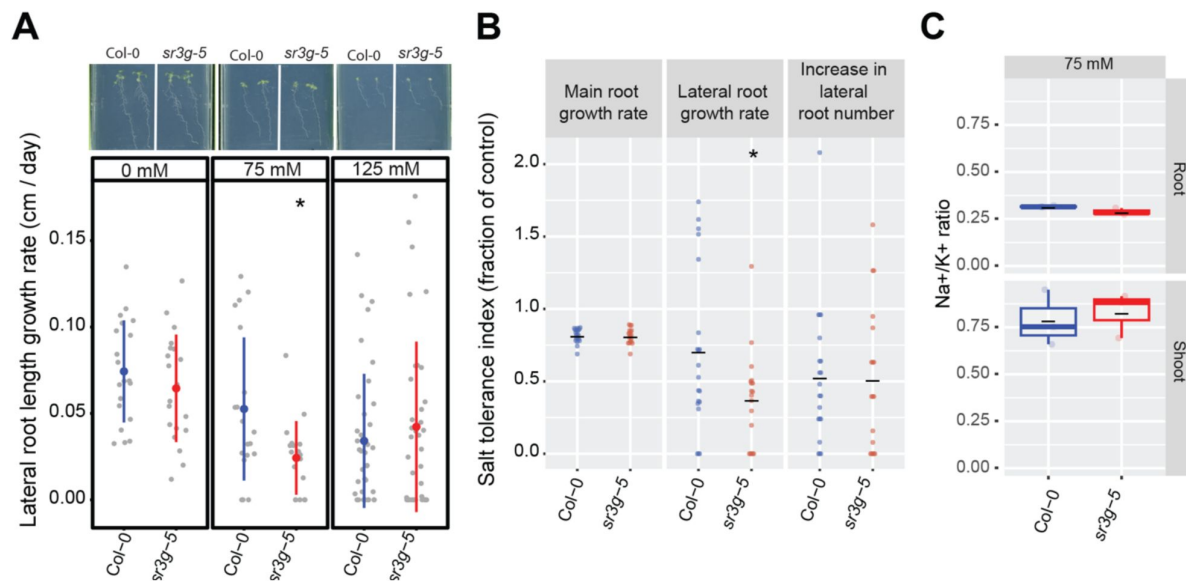


Figure 6.

***sr3g* mutant displays reduced growth rate for the lateral root length and Na⁺ content in the root.**

Root system architecture analysis of Col-0 and *sr3g-5* genotypes under various concentrations of NaCl are shown here. **(A)** Shown are representative images of 13-day-old Col-0 and *sr3g-5* genotypes that experienced 9 days of salt treatment at indicated concentrations as well as growth rate for lateral root length. **(B)** Salt Tolerance Index (STI) for the main root length, average lateral root length, and lateral root number at 75 mM NaCl. The STI was calculated by dividing the growth rate measured under salt stress by the growth rate measured under control condition for each genotype. **(C)** Na⁺/K⁺ ratio in root and shoot of Col-0 and *sr3g-5* after two weeks on 75 mM salt are shown. Each dot in (A) and (B) represents individual replicate per genotype. Lines in (A) and (B) graphs represent median and average, respectively. In (A) and (B), the asterisks above the graphs indicate significant differences between Col-0 and the *sr3g-5* mutant, as determined by the Student's t-test: **P* < 0.05. Statistical analysis in (C) was done by comparison of the means for all pairs using Tukey-Kramer HSD test. Levels not connected by the same letter are significantly different (*P* < 0.05). Root system architecture and ICP-AES analyses of Col-0 and *sr3g-5* mutant are shown in details in **Figure S17**.

was observed between Col-0 and *sr3g-5* mutant in terms of water content under either of conditions tested, providing no substantial evidence for SR3G to regulate water content (**Fig. S24**).

To investigate the role of SR3G in the maintenance of cellular integrity during salt stress, we evaluated the salt-induced changes in cell membrane integrity by measuring ion leakage in the leaf discs at the end of the experiment (**Fig. 7C**). While ion leakage increased in Col-0 under early salt stress conditions, no noticeable salt-induced change was observed for *sr3g-5* compared to control condition (**Fig. 7C**). Although the *sr3g-5* mutant exhibited significant increase in ion leakage in response to late salt stress, this phenomenon may be more closely associated with the onset of senescence rather than direct exposure to salt stress (**Fig. 7C**).

We further explored the role of SR3G in ion exclusion by quantifying the accumulation of Na⁺ and K⁺ ions in the rosette at the end of the experiment (4-weeks-old plants). We observed a reduced accumulation of Na⁺ in *sr3g-5* mutants compared to Col-0 for plants exposed to salt stress at an earlier timepoint (**Fig. 7D**). No significant differences in Na⁺ accumulation was observed between the Col-0 and *sr3g-5* at the late salt stress regime (**Fig. 7D**). A subtle, non-significant, reduction in K⁺ content was observed in *sr3g-5* mutant compared to Col-0 across all three conditions studied (**Fig. 7E**). In summary, these results suggest that SR3G plays a negative role in maintenance of plant growth, cellular integrity and Na⁺ accumulation during the salt stress, especially when the salt stress is applied during early vegetative development.

***sr3g* mutant displays enhanced root suberization**

As SR3G limits shoot growth, promotes cell damage and sodium translocation into shoot (**Fig. 7**), and is expressed in the most inner layers of the root (i.e., stele, **Fig. S13**), we hypothesized that SR3G plays a role in the regulation of suberization within the Casparian strips located in the endodermis, the layer adjacent to the stele. Salt stress exposure increases root suberization, thereby limiting Na⁺ loading into the transpiration stream and accumulation in the shoot (Baxter et al., 2009). We examined root suberization patterns in Col-0 and *sr3g-5* under non-stress and salt stress conditions. Root suberization was visualized using lipophilic stain Fluorol Yellow (FY), and staining intensity was examined under a confocal microscope. Additionally, we examined the developmental transition in root suberization through imaging four regions along the main root (**Fig. 8**). Visually, we observed that roots of *sr3g-5* were more suberized compared to Col-0 under both control and salt stress conditions (**Fig. 8A**). The quantification of FY signal intensity revealed non-significant, but substantially higher levels of root suberization in *sr3g-5* compared to Col-0 in sections one to three of the root under control condition (**Fig. 8B**). Only in root section four, the level of suberization significantly decreased in *sr3g-5* compared to Col-0 (**Fig. 8B**). In plants exposed to salt stress, we observed a non-significant difference between *sr3g-5* and Col-0 throughout the four sections, although, in section one which is the root tip proximal, *sr3g-5* showed more suberization compared to Col-0 (section 1, **Fig. 8A**). These results suggest that SR3G may play a negative role in root suberization.

To validate the FY histochemical analysis, we quantified root suberin monomers in the Arabidopsis seedlings using GC-MS analysis. Suberin monomers that were detected in the Arabidopsis roots included fatty acid (FA), fatty alcohol (OH-FA), α,ω -dicarboxylic acids (DCA), and ω -hydroxy fatty acid (ω OH-FA). Roots of *sr3g-5* contained significantly more total FA and DCA monomers compared to Col-0 under control condition, whereas the total OH-FA and ω OH-FA remained unchanged (**Fig. 8C**). In plants exposed to salt stress, the *sr3g-5* mutant showed an increased abundance of three suberin monomers, with DCA and ω OH-FA showing the most significant changes (**Fig. 8C**). The total FA decreased significantly in the mutant compared to Col-0 (**Fig. 8C**). Together, these results further corroborate with FY staining, suggesting that SR3G plays a negative role in root suberization.

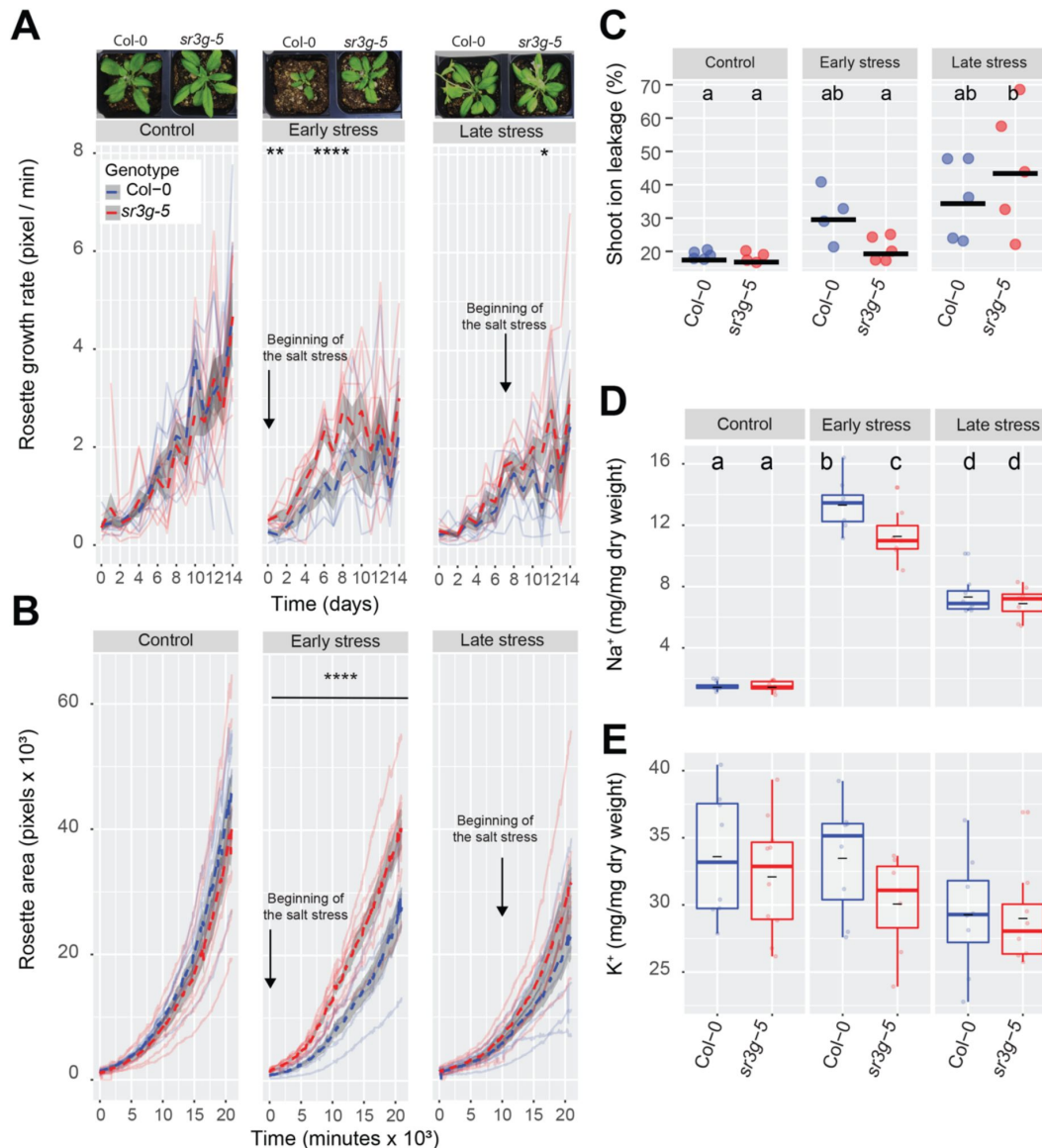


Figure 7.

***sr3g* mutant has higher growth rate and larger rosette area while displaying less ion leakage and Na⁺ accumulation in shoot under salt stress.**

The salt stress responses of 2-week-old (referred to as “early salt stress”) or 3-week-old (referred to as “late salt stress”) soil-grown Col-0 and *sr3g-5* plants that were exposed to a final concentration of 100 mM NaCl were examined here. Rosette growth rate (**A**) and area (**B**) were monitored over a period of two weeks, during which the “early salt stress” group experienced two weeks of salt stress while the “late salt stress” group experienced one week. Representative images of 4-week-old Col-0 and *sr3g-5* genotypes were shown in (A). Ion leakage percentage (**C**), Na⁺ (**D**) and K⁺ (**E**) contents in shoot were measured in 4-week-old plants. Each line in (A) and (B) represents the trajectory per genotype through time, where red and blue lines indicate Col-0 and *sr3g-5* plants grown under control or salt-stress conditions, respectively. The dashed lines represent the mean values of the genotype per condition, whereas the grey band represents the confidence interval. The asterisks above the graphs indicate significant differences between Col-0 and *sr3g-5* plants, as determined by one-way analysis of variance: **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001. Statistical analysis in (C) to (E) was done by comparison of the means for all pairs using Tukey–Kramer HSD test. Levels not connected by the same letter are significantly different (*P* < 0.05).

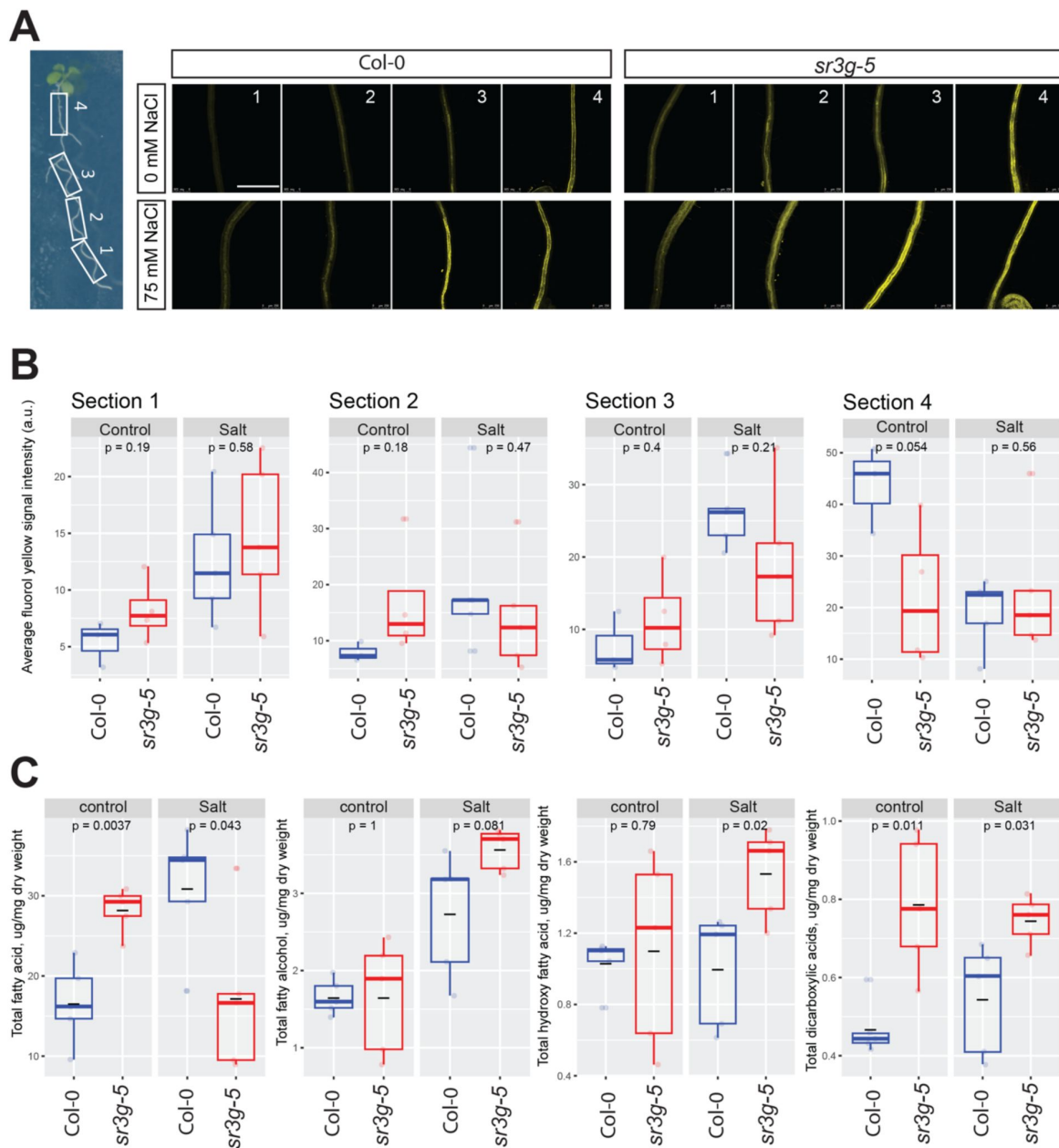


Figure 8.

***sr3g* mutant root is more suberized than Col-0.**

(A) Representative images of fluorol yellow (FY) staining of four root sections (# 1 to 4, from root tip to root base as shown by white rectangles across the seedling) were shown for Col-0 and *sr3g-5* mutant with or without 75 mM NaCl. Seeds were germinated on the 1/2 MS plates for 4 d and then seedlings were transferred to the treatment plates with or without 75 mM NaCl for 2 more days. (B) Quantification of FY signal intensity for each root section. The FY quantification was done using ImageJ. Scale bar is equal to 500 μ m in all. (C) Suberin monomers detected using Gas Chromatography–Mass Spectrometry (GC-MS) in the Col-0 and *sr3g-5* mutant roots with or without 75 mM NaCl included fatty acid (FA), fatty alcohol (OH-FA), α,ω -dicarboxylic acids (DCA), and ω -hydroxy fatty acid (ω OH-FA). The significant differences between Col-0 and *sr3g-5* mutant were evaluated using Student's t-test.

SR3G expression is regulated by WRKY75 transcription factor

To identify potential transcriptional regulators of *SR3G*, we examined the *SR3G* promoter for cis-regulatory elements using the CisCross tool (<https://plamorph.sysbio.ru/ciscross/>), and made a list of transcription factors that putatively interact with *SR3G* promoter (Fig. 9A). As the root stele showed the highest enrichment in the *SR3G* transcript (Fig. S13), we examined the transcription factors that were previously identified to be expressed in root stele (Brady et al., 2011). Cross-referencing of these two datasets revealed one transcription factor, *WRKY75* (AT5G13080) that was both enriched in root stele, and potentially binding to *SR3G* promoter.

WRKY75 was previously described to act as a positive regulator of salt stress response by enhancing *SOS1* expression (Lu et al., 2023). To investigate whether the *SR3G* expression depends on *WRKY75*, we examined the *SR3G* expression in the *wrky75* T-DNA insertion lines. *SR3G* expression was undetected in *wrky75* loss-of-function mutants (Fig. 9B), suggesting that *WRKY75* regulates *SR3G* expression. In comparison, expression of *WRKY75* remained unchanged in the *sr3g-5* mutant when compared to Col-0 (Fig. 9C). To investigate whether *WRKY75* directly binds to the *SR3G* promoter, we performed Y1H assay. Similar to the two positive controls, the strains transformed with *WRKY75*-AD and *SR3G*:pLacZ grew on the SD/-Trp/-Ura + X-Gal medium, thereby confirming the interaction between *WRKY75* and *SR3G* promoter region (Fig. 9D). In contrast, *WRKY75* did not show binding affinity to the promoter region of the adjacent gene to the *SR3G* (i.e., DUF247-150, AT3g50150), thereby reinforcing the uniqueness of the interaction between the *SR3G* and *WRKY75* (Fig. 9E).

To further examine the functional consequences of the relationship between *SR3G* and *WRKY75*, we generated *wrky75-3/sr3g-5* double mutants, and examined their root system architecture alongside the single mutants and Col-0. Under control conditions, both *sr3g-5* and *wrky75-3* mutants showed significant reduction in main root length compared to Col-0 and the two double mutants (Fig. 10A). Under 75 mM salt, only *wrky75-3* showed significant reduction in main root length compared to all other genotypes tested, confirming the previously described salt sensitivity for *wrky75* mutant under salt stress (Lu et al., 2023). We have not observed any differences in main root length under 125 mM salt across all genotypes tested (Fig. 10A). Regarding lateral root number (Fig. 10B) and length (Fig. 10C), we observed significant reduction in lateral root number and length for the two single mutants, *sr3g-5* and *wrky75-3*, compared to other genotypes only under control condition (Fig. 10B-C). We further investigated Na^+ and K^+ accumulations as well as Na^+/K^+ ratio in plate grown seedlings (Fig. 10D, Fig. S25, S26). At 75 mM salt concentration, a subtle reduction in root Na^+/K^+ ratio was observed in the *wrky75* single mutant compared to the other genotypes tested. Conversely, shoot Na^+/K^+ ratio exhibited a slight decrease in the two double mutants compared to the other three genotypes at 75 mM salt concentration. At 125 mM salt concentration, root Na^+/K^+ ratio decreased in the two double mutants compared to Col-0 and the two single mutants. Similarly, the two double mutants maintained lower Na^+/K^+ ratios compared to Col-0 and the two single mutants in shoot at 125 mM salt concentration (Fig. 10D), indicating that the *SR3G* mutation indeed compensated for the increased Na^+ accumulation observed in the *wrky75* mutant under salt stress.

Discussion

Exposure to abiotic stress alters plant architecture by affecting biomass distribution between individual organs (Guo et al., 2020; Jolkowska and Testerink, 2015). Exposure to drought alters the distribution of the biomass between root and shoot, to optimize water uptake, while limiting transpiration (Asch et al., 2005; Chen et al., 2022; Karcher et al., 2008; Uga et al., 2013). While the processes underlying salt-induced changes in shoot and root growth dynamics and architecture have been systematically described (Ariel et al., 2010; Awlia et al., 2021; Deolu-

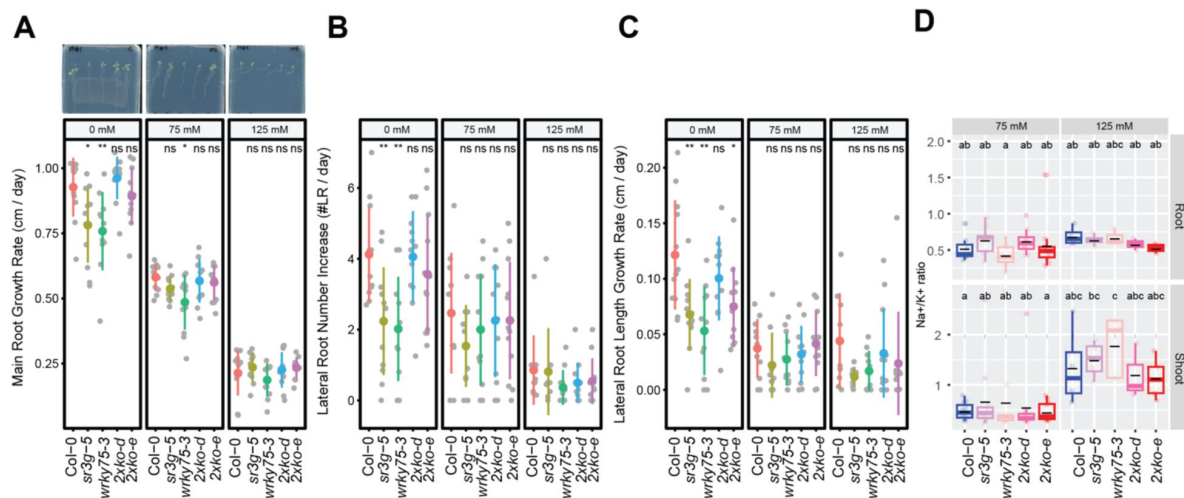


Figure 10.

***wrky75/sr3g* double mutants roots exhibit reduced sensitivity to salt stress and accumulate lower levels of Na⁺ in their shoots.**

(A-C) Root system architecture analysis for the *Col-0*, *sr3g-5*, *wrky75-3*, and two double mutants under various concentrations of NaCl are shown here. Main root growth rate (A), Lateral root number increase (B), and Lateral root length increase (C) are shown for the indicated genotypes at various salt concentrations. (D) Na⁺/K⁺ ratio in root and shoot of indicated genotypes after two weeks on treatment plates. Each dot represents individual replicate per genotype. The asterisks above the graphs (A-C) indicate significant differences between *Col-0* and the genotype, as determined by the Student's t-test: **P* < 0.05 and ***P* < 0.01. ns denotes no statistically significant. Statistical analysis in (D) was done by comparison of the means for all pairs using Tukey-Kramer HSD test. Levels not connected by the same letter are significantly different (*P* < 0.05).

Ajayi et al., 2019 [↗](#); Duan et al., 2013 [↗](#); Geng et al., 2013 [↗](#); Julkowska et al., 2017 [↗](#); Yutao Zou et al., 2022 [↗](#)), the methodological exploration of biomass distribution between root and shoot for plants exposed to salt stress is underexplored in the current body of research. Here, we developed a tool that allows a non-destructive quantification of root and shoot projected area of the seedlings grown on agar plates (**Fig. 1A** [↗](#)). This new tool allows us to efficiently describe changes in biomass accumulation and stress-induced alterations in root and shoot growth. Although the method is restricted to small plants and early developmental stages, it provides valuable insight into stress responses. The application of this tool to an Arabidopsis natural diversity panel revealed that salt stress exposure results in loss of coordination between root and shoot growth (**Fig. 1D** [↗](#)), rather than a clear directional change, as in the case of drought. To our knowledge, this is the first systematic study of changes in root:shoot ratio. The developed tool for quantifying root and shoot size in plate-grown Arabidopsis has the potential to significantly advance research by enabling developmental and stress studies, focusing on coordination of root and shoot development during seedling establishment. Facilitating detailed comparisons of root:shoot ratios, this tool can aid in formulating new hypotheses about the mechanisms contributing to seedling vigor and stress tolerance.

The natural variation in salt-induced changes in root:shoot ratio revealed a clear association with SNPs on chromosome 3 within *SR3G*, that is situated within a region containing eight other *DUF247* genes (**Fig. 2** [↗](#)). *SR3G* was previously associated with self-incompatibility in grasses (Crain et al., 2020 [↗](#); Lian et al., 2021 [↗](#)), as well as with lesions and dwarfism in Arabidopsis (Kondou et al., 2013 [↗](#)) through constitutive activation of defense responses. However, the *SR3G* associated with constitutive defense response was AT3G60470 (Kondou et al., 2013 [↗](#)), which is located outside the interval associated with salt stress-induced changes in root:shoot ratio (**Fig. 2C** [↗](#)). Because we did not observe any consistent phenotypes in the T-DNA insertion lines of *DUF247*s neighboring the *SR3G* (**Fig. S19-20**), we examined the RNAi lines targeting multiple *DUF247*s including *SR3G* (**Fig. S21-23**). Our root system architecture analyses revealed no redundancy between *SR3G* and other *DUF247*s, and the reduced lateral root development in double mutants of RNAi lines simply mimics the phenotype of *sr3g-5* single mutant (**Fig. 6** [↗](#), **Fig. S21**). Further, we demonstrated that only *SR3G* is regulated by *WRKY75* (**Fig. 9B-E** [↗](#)). As these genes are result of tandem duplication, they are expected to accumulate a lot of variation in the cis-regulatory regions. Therefore, differential regulation of two closely related genes can be expected by the accumulated variation within the promoter region.

The results acquired in this study provide multiple layers of evidence that support the role of *SR3G* as a negative regulator of salt stress response in Arabidopsis. Loss-of-function *SR3G* maintained higher shoot growth rate and larger rosette area when exposed to salt stress in the soil (**Fig. 7A-B** [↗](#)), probably due to reduced Na^+ accumulation (**Fig. 7D** [↗](#)), which in turn resulted in lower rates of cellular damage (**Fig. 7C** [↗](#)). This increased salt tolerance feature can be explained by the increased root suberization (**Fig. 7** [↗](#)) as suberin-deficient mutants show susceptibility to salt stress (Barberon et al., 2016 [↗](#); Shukla et al., 2021 [↗](#); Ursache et al., 2021 [↗](#)). For example, Arabidopsis *CYP86A1/HORST* (At5g58860) encodes a cytochrome P450-dependent enzyme that is responsible for catalyzing fatty acid ω -hydroxylation. It plays a pivotal role in the biosynthesis of aliphatic suberin, particularly expressed in the root endodermis. Knock-out mutations of *CYP86A1/HORST* result in a substantial decrease of approximately 60% in total root suberin (Höfer et al., 2008 [↗](#); Li et al., 2007 [↗](#)). Consequently, the diminished aliphatic suberin content in the roots of *cyp86a1* mutants leads to increased permeability to water and solutes (Ranathunge and Schreiber, 2011 [↗](#)). Interestingly, *SR3G* expression is enhanced by salt stress (**Figs. 4** [↗](#), **Fig. S13**), specifically in the root stele (**Fig. S13B**). The root stele encompasses vascular bundles, procambium and pericycle, which gives rise to the secondary growth and lateral root primordia respectively (Birnbaum et al., 2003 [↗](#)). The stele is central to the systemic transport of water, minerals, and nutrients within the plant (Li et al., 2017 [↗](#); Ramachandran et al., 2020 [↗](#); Zarebanadkouki et al., 2019 [↗](#)), and to the exclusion of salt ions from the transpiration stream (Møller et al., 2009 [↗](#)).

The deposition of hydrophobic residues of lignin and suberin in the Casparian strips poses a significant barrier to apoplastic transport (Foster and Miklavcic, 2017 [↗](#)). Previously, silencing genes involved in Casparian strips formation resulted in increased lateral root development (Wang et al., 2020 [↗](#)). As the suberization of the Casparian strips increases upon salt stress, it limits the accumulation of Na^+ and loss of K^+ ions from the root stele (Barberon et al., 2016 [↗](#)). However, the effects of increased root suberization under salt stress have thus far not been described to affect lateral root development. Here, we have identified *SR3G* as a negative regulator of root suberization under salt stress (Fig. 8 [↗](#)), which provides an additional layer of protection against sodium accumulation (Fig. 8 [↗](#)), but negatively impacts lateral root development (Fig. 6A-B [↗](#)).

As *SR3G* was previously not described in the context of salt stress, we attempted to identify its molecular context through its transcriptional regulation. Through our *in-silico* analysis, we have identified *WRKY75* as a putative regulator of *SR3G* expression, and confirmed it using RT-qPCR and Y1H assay (Fig. 9 [↗](#)). *WRKY75* was previously described to act as a positive regulator of salt stress tolerance through activation of *SOS1* expression (Lu et al., 2023 [↗](#)). Similar to previous study, we also observed *wrky75* mutants to be more sensitive to salt stress within our experiments (Fig. 10A-C [↗](#)). Nevertheless, when the *wrky75* mutation was coupled with *sr3g-5*, the roots displayed a phenotype identical to Col-0 (Fig. 10A [↗](#)), implying that the salt sensitivity of the *wrky75* mutant was counteracted by the *SR3G* mutation. Considering *WRKY75*'s role in positively regulating *SOS1* expression during salt stress, it is plausible that its binding to the *SR3G* promoter inhibits its expression, thereby enhancing salt tolerance on a larger scale.

One of the main players in Na^+ transport is High-affinity K^+ Transporter (*HKT1*) (Laurie et al., 2002 [↗](#)). *HKT1* is expressed in root stele (Mäser et al., 2002 [↗](#)), where it retrieves Na^+ from the transpiration stream and limits ion accumulation in the shoot, resulting in higher salt stress tolerance at later developmental stages (Møller et al., 2009 [↗](#)). At early developmental stages, *HKT1* overexpression reduces the development of lateral root primordia (Julkowska et al., 2017 [↗](#)), thus creating less Na^+ entry gate into the main root, resulting in less Na^+ accumulation in shoot. Similarly, *sr3g-5* mutants developed fewer lateral roots that are consistent with the lower Na^+ levels in the root upon salt stress (Fig. 6 [↗](#)). The effects of *HKT1* and *SR3G* differ in the magnitude of repression of lateral root development, with *HKT1* mutants resulting in almost complete cessation of lateral root development (Julkowska et al., 2017 [↗](#)), whereas repression of *SR3G* function has much more subtle phenotype. Whereas the effect of cell-type specific overexpression of *HKT1* on plant performance was negative during the early developmental stage, due to impairing lateral root development, the *sr3g* mutants showed increased salt tolerance when exposed to salt stress treatment during their early vegetative growth (Fig. 7 [↗](#)). These results suggest that reduced lateral root development under salt stress can have a gradient of responses leading to increased and decreased salt tolerance, and the effect is highly dependent on the wider physiological mechanisms, and their contributions to ion exclusion and maintained plant development throughout salt stress exposure.

In conclusion, the study of salt-induced changes in the root:shoot ratio, alongside the development of new tools to quantify these changes during the seedling establishment stage, represents an important new perspective into mechanisms underlying salt stress tolerance. Identification of *SR3G* as a novel regulator of root suberization, shoot growth, and Na^+ accumulation provides better understanding of molecular mechanisms underlying plant responses to salt stress, but also offers a new avenue to explore more nuanced and tissue-specific strategies for enhancing salt stress tolerance.

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Author contributions

M.M.J. and M.R.I. conceived the project and wrote the manuscript with inputs from all coauthors, C.T. and M.A.T. helped with conceiving the overall idea for the project, infrastructure and funding, R.A.H. and R.S. developed the tool to quantify root and shoot size from Arabidopsis grown on agar plates, A.K. assisted with GWAS analysis, M.R.I. conducted all experiments for functional characterization of *SR3G*, H.S., and M.D.A. performed molecular cloning of *DR3G*, H.S. screened other DUF247s T-DNA and RNAi mutant lines, L.J. performed the analysis of DUF247 domains that informed the cloning of individual *SR3G* fragments, G.D. and M.W. conducted GC-MS on suberin monomers, M.P. and E.C. performed ICP-AES, A.D.L.N. and L.Y. performed phylogenetic analysis, B.L. and Y.H. conducted Y1H.

Competing interests

The authors declare no competing interests **Correspondence and requests for materials** should be addressed to Dr. Magdalena M. Julkowska.

Supplemental Figure and Table Legends

Figure S1. Evaluation of the tool's precision for estimating seedling size. The tool used to quantify root and shoot size from the agar plates was tested on the Arabidopsis Col-0 seedlings exposed to various concentrations of salt stress (0, 75 and 125 mM NaCl). The fresh weight of root and shoot of individual plants were recorded at the last day of experiment (8 days after transfer to treatment plates, 12 days after germination), and the correlation analysis between individual organ's fresh weight and area was performed in R. The Correlation co-efficient and the p-value for individual correlations were calculated in R using stats package.

Figure S2. Salt stress reduced the increase in root and shoot area over time across HapMap accession. Four-days-old seedlings exposed to 0, 75 and 125 mM NaCl grown on agar plates were scanned every 2nd day and analyzed for root and shoot area using a custom tool. The change in genotype-specific **(A)** shoot and **(B)** root area was calculated in R using stats package. The thick dashed lines represent population-wide average in root and shoot area over the time of the experiment, while individual transparent lines indicate change in root and shoot area for individual accession. The differences between control and individual salt stress treatments were evaluated using ANOVA. The *, **, *** and **** represent p-values below 0.05, 0.01, 0.001 and 0.0001 respectively.

Figure S3. Salt stress induced changes in relative growth of shoot, root and root:shoot ratio. The stress tolerance index was calculated for shoot, root and root:shoot ratio by dividing genotype specific value recorded at salt stress over the genotypic mean of the trait recorded under non-stress conditions. Individual points represent individual genotypes. The population mean for 75 and 125 mM NaCl treated plants is represented by the colored point, whereas the population-wide standard error.

Figure S4. The correlation between all measured traits and calculated stress tolerance indices (STI). The correlations between all measured and calculated traits was performed using the Pearson's correlation coefficient. All non-significant correlations were replaced by blank circles. The size and color of individual circles indicates the strength and significance of individual correlation between two traits. Blue and red shades indicate positive and negative correlations respectively.

Figure S5. The correlation between total seedling area and salt-induced changes in the root-to-shoot ratio. The correlation between the total seedling area quantified using the custom tool at the last day of experiment (8 days after transfer and 12 days after germination), and the ratio of root-and-shoot growth rate was examined across Arabidopsis HapMap population and various concentrations of salt stress (0, 75 and 125 mM NaCl). The correlation analysis was performed in R, and the Correlation co-efficient (r) and the p -values for individual correlations were calculated in R using stats package.

Figure S6. Region around NTL8 is associated with salt stress-induced changes in root-per-shoot growth. The salt induced changes in root-to-shoot and shoot-per-total seedling area ratios were used as an input for GWAS. **(A)** Manhattan plot represents the associations of root-per-shoot growth factors recorded at 75 mM NaCl with 250 k SNPs while **(B)** represents the QQ plot for this association study. **(C)** Significant associations were found with the SNPs forming a locus on chromosome 2 in and around AT2G27300, encoding a NAC-domain containing transcription factor (NTL8). The natural variation in the LD region was studied in 162 accessions sequenced by 1001 Genomes Project. The upper panel represents portion of the missing data, upper middle panel represents deletions relative to Col-0, while lower middle panel represents the sequence similarity compared to Col-0. The bottom panel represents the open reading frames (ORFs) within the LD, and the location of associated SNPs is indicated with the dashed lines. Red dashed lines represent associations above Bonferroni threshold in 250k SNP mapping, while grey lines represent associations with $-\log_{10}(p\text{-value}) > 5$ in the 4M SNP mapping.

Figure S7. Region around unknown gene is associated with salt stress-induced changes in relative changes in shoot-per-total seedling area. The salt induced changes in root-to-shoot and shoot-per-total seedling area ratios were used as an input for GWAS. **(A)** Manhattan plot represents the associations of relative change (125 mM / 0 mM) in shoot-per-total seedling area at 4 days after transfer to 125 mM NaCl, while **(B)** represents the QQ plot for this association study. **(C)** Significant associations were found with the SNPs forming a locus on chromosome 3 in and around AT3G22430, encoding an unknown gene encoding a domain of unknown function. The natural variation in the LD region was studied in 162 accessions sequenced by 1001 Genomes Project. The upper panel represents portion of the missing data, upper middle panel represents deletions relative to Col-0, while lower middle panel represents the sequence similarity compared to Col-0. The bottom panel represents the open reading frames (ORFs) within the LD, and the location of associated SNPs is indicated with the dashed lines. Red dashed lines represent associations above Bonferroni threshold in 250k SNP mapping, while grey lines represent associations with $-\log_{10}(p\text{-value}) > 5$ in the 4M SNP mapping.

Figure S8. The QQ-plots for GWAS models with shoot-per-total seedling area. The QQ-plots for association between shoot-per-total seedling area recorded 6 days after transfer to 125 mM NaCl and **(A)** 250 k SNPs and **(B)** 4 M SNPs. The red line indicates the expected vs. observed ratio of

association, while black points indicate the observed associations.

Figure S9. Haplotype analysis of AT3G50160 (SR3G). The haplotype analysis performed using SNPs located within the coding region of the SR3G (AT3G50160) revealed significant differences in **(A)** Relative shoot growth (as fraction of shoot growth at 125 mM NaCl and 0 mM NaCl) and **(B)** Shoot growth at 125 mM NaCl. The significant differences between individual haplotype groups (each haplotype represented by at least 3 accessions), were tested using ANOVA with Tukey HSD to identify significantly different groups. Significant differences were identified between Haplotype groups 1 (represented by 46 accessions, including Col-0) and 11 (represented by 11 accessions, including Blh-1).

Figure S10. Nucleotide sequence alignment of SR3G CDS between Col-0 and Blh-1 alleles. The nucleotide sequences from SR3G CDS were obtained from Col-0 and Blh-1 cDNA, and compared to Col-0 cDNA sequence on TAIR. The cDNA sequences were subsequently aligned in JalView using ProbconsWS alignment. The intensity of purple hue indicates the nucleotide identity, while dash marks indicate insertion in Blh-1 compared to Col-0 allele.

Figure S11. Amino acid sequence alignment of SR3G CDS between Col-0 and Blh-1 alleles. The amino acid sequences from SR3G CDS were obtained by translating Col-0 and Blh-1 cDNA into amino acid sequence in JalView using Standard settings. The amino acid sequences were subsequently aligned in JalView using ProbconsWS alignment. The sequence length markers for are indicated above the aligned sequence. The intensity of purple hue is indicating the nucleotide identity, while dash marks indicate insertion in one of the alleles. * indicate STOP codons.

Figure S12. Sequence alignment between Col-0 and Blh-1 alleles for SR3G promotor. The nucleotide sequences from SR3G promotor were obtained from Col-0 and Blh-1 genomic DNA, and compared to Col-0 sequence on TAIR. The gDNA sequences were subsequently aligned in JalView using ProbconsWS alignment. The intensity of purple hue is indicating the nucleotide identity, while dash marks indicate insertions.

Figure S13. Gene expression profile of SR3G across various developmental stages and salt stress. The absolute gene expression of *SR3G* is depicted across various developmental stages **(A)**, root cell types **(B)**, and in response to salt-induced changes **(C)**. The images were acquired through the eFP Browser using data sources specified for the developmental map, abiotic stress, and tissue specific.

Figure S14. Gene expression profile of other DUF247s under salt stress. Absolute gene expression of other *DUF247s* are depicted in response to salt-induced changes. The images were acquired through the eFP Browser using data sources specified for the abiotic stress. The *SR3G* is shown in a green box for comparison.

Figure S15. SR3G predicted protein domains. The SR3G protein sequence (Col-0 allele) was used to identify predicted protein domains. The identified domains include: nuclear localization signal (NLS): K15-F19; coiled-coil domain: M153-K173; potential Ca²⁺ binding pocket: G210-F372; and transmembrane domain P469-T503. Informed by these predicted protein domains, we designed SR3G Fragment A, B and C for subsequent localization studies and functional characterization. The pink mark at Y216 indicates a position of a premature STOP codon in Blh-1 allele.

Figure S16. The expression of DUF247-150 remains unaltered in the *sr3g* mutant. **(A)** Schematic diagram of main *SR3G* gene model and the location of T-DNA insertions. The location of T-DNA insertions is shown for *sr3g-4*, and -5 mutants using triangles. RT-qPCR showing expression of *SR3G* gene (AT3G50160) **(B)** and its neighboring gene, *DUF247-150* (At3g50150) **(C)** in Col-0 and two *sr3g* mutants. RT-qPCR analyses were conducted using seedlings grown on 1/2x MS for 4 d and then followed by transferring to the treatment plates with or without 75 mM NaCl for one week.

Mean values are shown $\pm$ SE, with the number of replicates (n) shown in each graph. *AT4G04120* (transposable_element_gene) was used as reference gene for normalization. Significance was determined by the Tukey–Kramer HSD test in JMP. Levels not connected by same letter are significantly different.

Figure S17. *sr3g-5* mutant displays no difference in main root length and lateral root numbers compared to Col-0. Root system architecture analysis of Col-0 and *sr3g-5* genotypes under various concentrations of NaCl are shown here for main root length (A) and increase in lateral root number (B). 13-day-old Col-0 and *sr3g-5* genotypes experienced 9 days of salt treatment at indicated concentrations were used here. Each dot in (A) and (B) represents individual replicate per genotype. Statistical analysis was performed between Col-0 and the *sr3g-5* mutant using the Student's t-test: $*P < 0.05$. (C) Na^+ , (D) K^+ contents in root and shoot of Col-0 and *sr3g-5* after two weeks on 75 mM salt are shown. Statistical analysis in (C–D) was done by comparison of the means for all pairs using Tukey–Kramer HSD test. Levels not connected by the same letter are significantly different ($P < 0.05$).

Figure S18. *sr3g-4* mutant has no alteration in root system architectures compared to Col-0. Root system architecture analysis of Col-0 and *sr3g-4* genotypes under various concentrations of NaCl are shown here for main root length (A), lateral root length (B), and lateral root number (C). Salt Tolerance Index (STI) for the main root length, average lateral root length, and lateral root number are shown at 75 (D) and 125 (E) mM NaCl, respectively. The STI was calculated by dividing the growth rate measured under salt stress by the growth rate measured under control condition for each genotype. Statistical analysis was performed between Col-0 and the *sr3g-4* mutant using the Student's t-test: $*P < 0.05$.

Figure S19. Root system architecture analysis of neighboring DUF247 mutants under the salt. Main root length (A), lateral root number (B), and lateral root length (C) are shown for neighboring *DUF247*s of the *SR3G*, including At3g50120 (shown as 120, with one T-DNA representative, 120-2(SAIL_382_A09)), At3g50140 (shown as 140, with three T-DNA representatives, 140-1 (SALK_044685), 140-2 (SALK_005466), and 140-3 (SALK_122700)), At3g50150 (shown as 150, with two T-DNA representatives, 150-1 (SALK_003824) and 150-3 (SALK_071080)), and At3g50170 (shown as 170, with two T-DNA representatives, 170-1 (SALK_009186) and 170-2 (SALK_112602)) compared to Col-0 under various concentrations of NaCl as indicated. The corresponding Salt Tolerance Index (STI) was calculated for main root growth (D), increase in lateral root number (E), and lateral root length (F). Statistical analysis was performed between Col-0 and the other mutants using the Student's t-test: $*P < 0.05$. The root system architecture analysis was performed similar to what described for [Figure 6](#).

Figure S20. Root system architecture analysis of neighboring DUF247 mutants under the salt, continues. Main root length (A), lateral root number (B), and lateral root length (C) are shown for neighboring of the *SR3G*, including At3g50170 (shown as 170, with three T-DNA representatives, 170-3F (SALK_008710), 170-4F (SALK_145999), 170-5A (SALK_072937)), At3g50190 (shown as 190, with one T-DNA representative, 190-2B (SALK_137791)), and At3g50200 (shown as 200, with three T-DNA representatives, 200-1A (SALK_113759), 200-2B (SALK_093701), and 200-3B (SALK_129634)) compared to Col-0 under various concentrations of NaCl as indicated. The corresponding Salt Tolerance Index was calculated for main root growth (D), increase in lateral root number (E), and lateral root length (F). Statistical analysis was performed between Col-0 and the other mutants using the Student's t-test: $*P < 0.05$. The root system architecture analysis was performed similar to what described for [Figure 6](#).

Figure S21. Root system architecture analysis of RNAi lines targeting neighboring DUF247s. Main root length (A), main root length tolerance index (B), lateral root length (C), lateral root length tolerance index (D), lateral root number (E), and lateral root number tolerance index (F) are shown for RNAi lines targeting AT3G50140 (CSHL3-F1a), targeting AT3G50130 and AT3G50140

(CSHL6-C2d), targeting AT3G50150 and AT3G50170 (CSHL11-F1b), targeting AT3G50140 and AT3G50170 (CSHL13-F2a), targeting AT3G50150 and AT3G50190 (CSHL18-K4c), targeting AT3G50130, AT3G50140, AT3G50150, AT3G50160, AT3G50170, AT3G50180, and AT3G50190 (CSHL19-E2c), targeting AT3G50130 and AT3G50150 (CSHL21-E2a), and targeting AT3G50150 and AT3G50160 (CSHL22-F1c). Statistical analysis was performed between Col-0 and the RNAi lines using a Student's t-test: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

Figure S22. Root system architecture analysis of RNAi lines targeting neighboring DUF247s continued. Main root length (A), main root length tolerance index (B), lateral root length (C), lateral root length tolerance index (D), lateral root number (E), and lateral root number tolerance index (F) are shown for RNAi lines targeting AT3G50160 (CSHL4-B2a and CSHL4G1d), targeting AT3G50140 and AT3G50150 (CSHL7-A3a), targeting AT3G50140 and AT3G50190 (CSHL8-G2b), targeting AT3G50180 (CSHL10-I1c), targeting AT3G50130, AT3G50140, AT3G50170, AT3G50180, and AT3G50190 (CSHL12-A1a), targeting AT3G50140, AT3G50170, AT3G50180, and AT3G50190 (CSHL15-E1b), targeting AT3G50130, AT3G50160, and AT3G50170 (CSHL-A2c and CSHL16-B1a), targeting AT3G50130 and AT3G50160 (CSHL20-G3b), and targeting AT3G50130 and AT3G50150 (CSHL21-L2c and CSHL21-M1d). Statistical analysis was performed between Col-0 and the RNAi lines using a Student's t-test: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

Figure S23. Root system architecture analysis of RNAi lines targeting neighboring DUF247s continued. Main root length (A), main root length tolerance index (B), lateral root length (C), lateral root length tolerance index (D), lateral root number (E), and lateral root number tolerance index (F) are shown for RNAi lines targeting AT3G50140 (CSHL3-F3a), targeting AT3G50160 (CSHL4-D1c3, CSHL4-D4a, and CSHL4-F1b), targeting AT3G50130 and AT3G50170 (CSHL5-D3b and CSHL5-E2b), targeting AT3G50130 and AT3G50140 (CSHL6-B1c), targeting AT3G50140 and AT3G50150 (CSHL7-C1b2), targeting AT3G50140 and AT3G50190 (CSHL8-D3b and CSHL8-H1d), targeting AT3G50180 (CSHL10-I1b), targeting AT3G50150 and AT3G50170 (CSHL11-E5b), targeting AT3G50130, AT3G50140, AT3G50170, AT3G50180, and AT3G50190 (CSHL12-B1a and CSHL12-D1c), targeting AT3G50140 and AT3G50170 (CSHL13-F1b and CSHL13-F4d), targeting AT3G50140, AT3G50170, AT3G50180, and AT3G50190 (CSHL15-G1a1 and CSHL15-H2b), targeting AT3G50150 and AT3G50190 (CSHL18-E4c and CSHL18-I2d), targeting AT3G50130, AT3G50140, AT3G50150, AT3G50160, AT3G50170, AT3G50180, and AT3G50190 (CSHL19-F1a, CSHL19-I3c, and CSHL19-L1c), and targeting AT3G50130 and AT3G50160 (CSHL20-F2b). Statistical analysis was performed between Col-0 and the RNAi lines using a Student's t-test: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

Figure S24. Plant water content comparison shows no significant differences between sr3g mutant and Col-0 wild-type plants. Plant water content was calculated by subtracting rosette dry weight from fresh weight for the 2-week-old (referred to as “early salt stress”) or 3-week-old (referred to as “late salt stress”) soil-grown Col-0 and sr3g-5 plants that were exposed to a final concentration of 100 mM NaCl. “early salt stress” group experienced two weeks of salt stress while the “late salt stress” group experienced one week. Statistical analysis was done by comparison of the means for all pairs using Tukey–Kramer HSD test. Levels not connected by the same letter are significantly different ($P < 0.05$).

Figure S25. *wrky75/sr3g* double mutants exhibit higher shoot K^+ contents compared to Col-0 and each individual single mutant in the presence of 125 mM salt. (A) Na^+ and (B) K^+ contents in root and shoot of Col-0, *sr3g-5*, *wrky75-3*, and two double mutants under various concentrations of NaCl are shown here. Statistical analysis in (A-B) was done by comparison of the means for all pairs using Tukey–Kramer HSD test. Levels not connected by the same letter are significantly different ($P < 0.05$).

Figure S26. *wrky75/sr3g* double mutants exhibit no noticeable differences in rosette area compared to their individual single mutants when grown in saline soil. The salt stress responses of 2-week-old soil-grown plants that were exposed to a final concentration of 100 mM NaCl were examined here. Rosette area (A) and growth rate for each individual day were shown (B-M). Growth rate is shown for 1 (B), 2 (C), and 3 (D) days before the stress, respectively. Growth rate is shown for 1 (E), 2 (F), 3 (G), 4 (H), 5 (I), 6 (J), 7 (K), 8 (L), and 9 (M) days of stress application, respectively. Na⁺/K⁺ (N) ratio in shoot were measured in 4-week-old plants. Each line in (A) represents the trajectory per genotype through time. The effect of the genotype within individual treatment in (B-M) was tested using ANOVA, and the significantly different genotypes were additionally determined using Tukey HSD test (pvalue < 0.05). Statistical analysis in (N) was done by comparison of the means for all pairs using Tukey–Kramer HSD test. Levels not connected by the same letter are significantly different ($P < 0.05$).

Table S1. The individual traits recorded and calculated from the HapMap accessions exposed to the three different levels of salt stress. The root and shoot projected area were quantified using a custom developed tool. The change in the root and shoot area over time was used to calculate Growth Factor (GF) by fitting an exponential function. The broad sense heritability (H₂) was calculated for each trait over individual days and conditions using MVApp.

Table S2. Genotypic mean data used as an input for GWAS. The genotypic mean data was calculated from at least 4 biological replicates per genotype per condition. The root and shoot area were extracted using a custom developed tool. The Growth Factors (GF) for root and shoot were calculated using exponential growth function. Total seedling size (Tot) was calculated by adding root and shoot area for individual day. Shoot per Total (SpT) ratio was calculated for individual days by dividing genotype specific shoot area by the total seedling area for that day and condition. Root per Shoot ratio (RpS) was calculated by dividing genotype specific root area by the shoot area for that day and condition.

Table S3. Significant associations identified with GWAS of all used traits. The location of individual SNPs is listed according to its location on Chromosome (Chr) and position (Pos). The individual SNP Minor Allele Count (MAC) and Minor Allele Frequency (MAF) is calculated based on specific SNP set used (4 Million or 250 Thousand – listed in “Mapping” column). The traits associated with individual SNPs are abbreviated as RpS for Root-per-Shoot, SpT for Shoot-per-Total seedling area, Tot for Total seedling area, GR for growth rate and SHIT for Shoot Ion Independent Tolerance index (Salt / Control).

Table S4. Most important GWAS associations. The most important associations identified through GWAS (Table S3) were inspected further for their functions using either T-DNA or RNAi mutant lines. List of oligoes used for genotyping, RT-qPCR, cloning, and Y1H assay are provided.

Table S5. The gene ID, species, and gene names used for identification of SR3G orthologs and paralogs.

Table S6. Background and foreground values of omega.

Table S7. Links to cloning construct maps generated in this study.

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Reviewer #1 (Public review):

Summary:

The authors aim to assess the effect of salt stress on root:shoot ratio, identify the underlying genetic mechanisms, and evaluate their contribution to salt tolerance. To this end, the authors systematically quantified natural variations in salt-induced changes in root:shoot ratio. This innovative approach considers the coordination of root and shoot growth rather than exploring biomass and the development of each organ separately. Using this approach, the authors identified a gene cluster encoding eight paralog genes with a domain-of-unknown-function 247 (DUF247), with the majority of SNPs clustering into SR3G (At3g50160). In the manuscript, the authors utilized an integrative approach that includes genomic, genetic, evolutionary, histological, and physiological assays to functionally assess the contribution of their genes of interest to salt tolerance and root development.

Comments on revisions:

As the authors correctly noted, variations across samples, genotypes, or experiments make achieving statistical significance challenging. Should the authors choose to emphasize trends across experiments to draw biological conclusions, careful revisions of the text, including titles and figure legends, will be necessary to address some of the inconsistencies between figures (see examples below). However, I would caution that this approach may dilute the overall impact of the work on SR3G function and regulation. Therefore, I strongly

recommend pursuing additional experimental evidence wherever possible to strengthen the conclusions.

(1) Given the phenotypic differences shown in Figures S17A-B, 10A-C, and 6A, the statement that "SR3G does not play a role in plant development under non-stress conditions" (lines 680-681) requires revision to better reflect the observed data.

(2) I agree with the authors that detecting expression differences in lowly expressed genes can be challenging. However, as demonstrated in the reference provided (Lu et al., 2023), a significant reduction in WRKY75 expression is observed in T-DNA insertion mutant alleles of WRKY75. In contrast, Fig. 9B in the current manuscript shows no reduction in WRKY75 expression in the two mutant alleles selected by the authors, which suggests that these alleles cannot be classified as loss-of-function mutants (line 745). Additionally, the authors note that the *wrky75* mutant exhibits reduced main root length under salt stress, consistent with the phenotype reported by Lu et al. (2023). However, other phenotypic discrepancies exist between the two studies. For example, 1) Lu et al. (2023) report that *w-rky75* root length is comparable to WT under control conditions, whereas the current manuscript shows that *wrky75* root growth is significantly lower than WT; 2) under salt stress, Lu et al. (2023) show that *wrky75* accumulates higher levels of Na⁺, whereas the current study finds Na⁺ levels in *wrky75* indistinguishable from WT. To confirm the loss of WRKY75 function in these T-DNA insertion alleles the authors should provide additional evidence (e.g., Western blot analysis).

<https://doi.org/10.7554/eLife.98896.2.sa2>

Reviewer #2 (Public review):

Summary:

Salt stress is a significant and growing concern for agriculture in some parts of the world. While the effects of sodium excess have been studied in *Arabidopsis* and (many) crop species, most studies have focused on Na uptake, toxicity and overall effects on yield, rather than on developmental responses to excess Na, per se. The work by Ishka and colleagues aims to fill this gap.

Working from an existing dataset that exposed a diverse panel of *A. thaliana* accessions to control, moderate, and severe salt stress, the authors identify candidate loci associated with altering the root:shoot ratio under salt stress. Following a series of molecular assays, they characterize a DUF247 protein which they dub SR3G, which appears to be a negative regulator of root growth under salt stress.

Overall, this is a well-executed study which demonstrates the functional role played by a single gene in plant response to salt stress in *Arabidopsis*.

Review of revised manuscript:

The authors have addressed my point-by-point comments to my satisfaction. In the cases where they have changed their manuscript language, clarified figures, or added analyses I have no further comment. In some cases, there is a fruitful back-and-forth discussion of methodology which I think will be of interest to readers.

I have nothing to add during this round of review. I think that the paper and associated discussion will make a nice contribution to the field

<https://doi.org/10.7554/eLife.98896.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors aim to assess the effect of salt stress on root:shoot ratio, identify the underlying genetic mechanisms, and evaluate their contribution to salt tolerance. To this end, the authors systematically quantified natural variations in salt-induced changes in root:shoot ratio. This innovative approach considers the coordination of root and shoot growth rather than exploring biomass and the development of each organ separately. Using this approach, the authors identified a gene cluster encoding eight paralog genes with a domain-of-unknown-function 247 (DUF247), with the majority of SNPs clustering into SR3G (At3g50160). In the manuscript, the authors utilized an integrative approach that includes genomic, genetic, evolutionary, histological, and physiological assays to functionally assess the contribution of their genes of interest to salt tolerance and root development.

Strengths:

The holistic approach and integrative methodologies presented in the manuscript are essential for gaining a mechanistic understanding of a complex trait such as salt tolerance. The authors focused on At3g50160 but included in their analyses additional DUF247 paralogs, which further contributes to the strength of their approach. In addition, the authors considered the developmental stage (young seedlings, early or late vegetative stages) and growth conditions of the plants (agar plates or soil) when investigating the role of SR3G in salt tolerance and root or shoot development.

Weaknesses:

The authors' claims and interpretation of the results are not fully supported by the data and analyses. In several cases, the authors report differences that are not statistically significant (e.g., Figures 4A, 7C, 8B, S14, S16B, S17C), use inappropriate statistical tests (e.g., t-test instead of Dunnett Test/ANOVA as in Figures 10B-C, S19-23), present standard errors that do not seem to be consistent with the post-hoc Tukey HSD Test (e.g., Figures 4, 9B-C, S16B), or lack controls (e.g., Figure 5C-E, staining of the truncated versions with FM4-64 is missing).

We thank the reviewer for their critical thoughts on the presented data. We have revised our data interpretation in the main text to more accurately reflect the results. Given the nature of our experimental setup, where we trace the roots of individual Arabidopsis seedlings grown on plates, there is considerable biological variation, which makes achieving strong statistical significance between samples or genotypes challenging. However, we think that the representation of the data as transparently as possible is necessary to provide the readers and reviewers a true picture of the variability that we are observing. Consequently, we have centered our data interpretation around observable trends that facilitate drawing conclusions.

The choice of statistical test is closely tied to the specific biological question being addressed. In Figures 10A-C, as in Figures 6A-B, we compared all genotypes to the wild-type Col-0 within each condition, and thus ANOVA analysis, testing the general effect of the genotype across

both mutants and Col-0 wild-type is not appropriate. Similarly, in Figures S19-S23, we compared each mutant line to the wild-type Col-0 under each condition.

We repeated the post-hoc Tukey HSD Test for Figures 4, 9B-C, and S16B and made adjustments where necessary (see tracked changes manuscript).

The truncated versions do not localize to the plasma membrane; instead, they are targeted to the nucleus and cytosol, mimicking the localization pattern of free GFP, which was used as a control in Panel F. Therefore, we believe that having FM4-64 as a control for these specific images is not informative, but instead using free GFP is serving as a better control in that particular construct.

In other cases, traits of root system architecture and expression patterns are inconsistent between different assays despite similar growth conditions (e.g., Figures S17A-B vs. 10A-C vs. 6A, and Figures S16B vs. 4A/9B), or T-DNA insertion alleles of WRKY75 that are claimed to be loss-of-function show comparable expression of WRKY75 as WT plants. Additionally, several supplemental figures are mislabeled (Figures S6-9), and some figure panels are missing (e.g., Figures S16C and S17E).

We thank the reviewer for raising these points and noticing the inconsistency between different assays (e.g., Figures S17A-B vs. 10A-C vs. 6A, and Figures S16B vs. 4A/9B). As mentioned above, considerable biological variation makes achieving strong statistical significance between samples, genotypes, or experiments challenging. Thus, we have centered our data interpretation around observable “trends” between experiments to facilitate drawing conclusions. Considering Figures S17A-B, 10A-C, and 6A, we acknowledge the reviewer’s concern about inconsistencies in root system architecture across experiments. Initially, we observed that the *sr3g* mutant had reduced lateral root length compared to Col-0 under salt stress. This led us to focus on this specific phenotypic trait rather than the overall root system architecture. Despite some variation, the *sr3g* mutant consistently showed a similar trend/phenotype when compared to Col-0 under salt stress. We believe the variation in main root length and lateral root number between experiments is due to inherent differences between biological replicates.

Regarding gene expression patterns between Figures S16B and 4A/9B, we included part of Figure 9B (SR3G gene expression in Col-0) in Figure 4A. Figure S16B represents a completely different assay. Despite variations between assays, the overall message remains consistent: SR3G gene expression is induced under salt stress in the root but not in the shoot.

Both SR3G and WRKY75 are expressed at very low levels, even under the 75 mM salt stress condition we tested. When gene expression is so low, detecting changes is challenging due to inherent variations. Nonetheless, we observed a reduction in WRKY75 expression in the mutant lines compared to wild-type Col-0, though this reduction was not statistically significant. More importantly, we observed a similar phenotype in the *wrky75* mutant, specifically reduced main root length under salt stress, consistent with the findings of the published paper in *The Plant Cell* by Lu et al. (2023) “Lu, K.K., Song, R.F., Guo, J.X., Zhang, Y., Zuo, J.X., Chen, H.H., Liao, C.Y., Hu, X.Y., Ren, F., Lu, Y.T. and Liu, W.C., 2023. CycC1; 1–WRKY75 complex-mediated transcriptional regulation of SOS1 controls salt stress tolerance in *Arabidopsis*. *The Plant Cell*, 35(7), pp.2570-2591”.

We appreciate the reviewer for spotting the missing labels for Figures S6-9. We corrected them at the main text, figures, and legends. We added panel C to Figure S16 and removed panel E from Figure S17 legend, now they match to actual figures and legends.

Consequently, the authors' decisions regarding subsequent functional assays, as well as major conclusions about gene function, including SR3G function in root system

architecture, involvement in root suberization, and regulation of cellular damage are incomplete.

We greatly appreciate the reviewer's thorough review of our manuscript and their critical comments. We have carefully addressed all comments and concerns.

Reviewer #2 (Public Review):

Salt stress is a significant and growing concern for agriculture in some parts of the world. While the effects of sodium excess have been studied in Arabidopsis and (many) crop species, most studies have focused on Na uptake, toxicity, and overall effects on yield, rather than on developmental responses to excess Na, per se. The work by Ishka and colleagues aims to fill this gap.

Working from an existing dataset that exposed a diverse panel of A. thaliana accessions to control, moderate, and severe salt stress, the authors identify candidate loci associated with altering the root:shoot ratio under salt stress. Following a series of molecular assays, they characterize a DUF247 protein which they dub SR3G, which appears to be a negative regulator of root growth under salt stress.

Overall, this is a well-executed study that demonstrates the functional role played by a single gene in plant response to salt stress in Arabidopsis.

The abstract and beginning of the Discussion section highlight the "new tool" developed here for measuring biomass accumulation. I feel that this distracts from the central aims of the study, which is really about the role of a specific gene in root development under salt stress. I would suggest moving the tool description to less prominent parts of the manuscript.

We appreciate the reviewer's suggestion. We believe that the innovative tool used to extract shoot-to-root ratio data from previous experiments underscores the value of reutilizing previously acquired data for new discoveries and demonstrates how reanalyzing the same data can provide fresh insights, such as identification of new allelic variation. Therefore, we decided to retain this section, as our discovery of the SR3G gene originated from this innovative tool.

Recommendations for the authors:

Reviewer #3 (Recommendations For The Authors):

Line 58 (opening sentence) - salt accumulation in the soil is not caused by evaporation exceeding input; that scenario results in soil water deficit. The issue is when the input water has dissolved ions.

We thank the reviewer for raising this important point. While this point is theoretically true, all of the water that is found in natural environments contains some dissolved ions. Therefore, drought conditions will lead, over time, to increased soil salinization. We have amended this sentence to represent our point better.

“Salt stress is predominant in the dryland areas where evaporation rate exceeds water input. As all water contains dissolved ions, the prolonged exposure to drought stress results in increased accumulation of salts in the upper soil layers 1–3.”

I feel that it would be helpful, for replication and for interpretation, if the authors could provide water potentials for the growing media used throughout. What water potentials are the plants experiencing when grown in 1/2 MS + agar at 0, 75, and 150mM NaCl?

Juenger and Verslues present a great recent discussion of the importance of reporting these values (Juenger, T. E. and P. E. Verslues (2023). "Time for a drought experiment: Do you know your plants' water status?" Plant Cell 35(1): 10-23.)

Critically, how do the water potentials experienced by agar-grown plants compare to those experienced in soil-grown plants? As a stated aim of this study is to allow translation to crops these data are very important to convince physiologists of the relevance of the results.

We thank the reviewer for raising this important point. We completely agree that growing plants on agar plates is an artificial setup and knowing the water potential of the plants within this setup would be highly informative. However, as indicated in review by Juenger and Verslues 2023, the agar plate setup is much more reproducible compared to various soil conditions, and we report the media composition in sufficient detail for it to be reproduced in other laboratory conditions.

Furthermore, while investigating the water status of plants and soil is indeed intriguing, it is beyond the scope of this study and would require us to redo the experiments with specific tools listed within the Juenger and Verslues review, which are currently not within our laboratory equipment list.

Importantly, any changes reported in this manuscript apply equally to both wild-type and mutant lines under all conditions. We provide extensive report on the soil type used, as well as soil quantity. We are using the gravimetric method to determine the water content, and salt stress application, as described in previous works from our lab (Yu and Sussman et al., 2024 Plant Physiology and Awlia et al., 2016 Frontiers in Plant Science).

Nonetheless, we have now included water content measurements for soil-grown plants under different conditions, calculated by subtracting dry weight from fresh weight (new Fig. S24). Although plant water content may not fully capture the water status of the media or soil, our measurements did not reveal any significant differences in water content between genotypes across the various conditions tested.

Line 69- missing an "and" after "(ABA)."

Thanks. We added the missing "and".

Line 79 - I think the association being made is between natural variation in root and shoot growth and genetic variants, not "underlying genes."

We thank the reviewer for this suggestion. The cause for the identified association indeed relies on allelic variation within the genetic region. We have re-phrased this sentence within the manuscript.

"Many forward genetic studies were highly successful in associating natural variation in root and shoot growth with allelic variation in gene coding and promoter regions, thereby identifying potential new target traits for improved stress resilience 18,20,21."

Figure 1 - what do "seGF" and "reGF" stand for? Shoot and root growth rate, respectively, but there are extra letters in there...

The abbreviations stand for shoot exponential Growth Factor and root exponential Growth factor. An explanation of the acronym has been added to the text.

"The increase in the projected area of shoot and root (Fig. S2) was used to estimate (A) shoot and (B) root exponential growth rate (seGR and reGR respectively)."

Figure 1 legend - there's an "s" missing in "across." And two "additionally" in the penultimate sentence.

Thanks for spotting the errors. We fixed these errors.

Line 109 - how was the white balance estimated for the images on the flatbed scanner?

Within the developed tool, we have not adjusted or controlled for white balance in any way, as the white balance from the flatbed scanner is kept at one value. The tool transforms the imaged pixels into bins consisting of white (root), green (shoot), and blue (place) pixels based on the closest distance in the RGB scale to the particular color, which makes correcting for white balance obsolete. We have provided an additional explanation for this within the M&M section.

“A Matlab-based tool was developed to simplify and speed up the segmentation and analysis pipeline. For automatic segmentation, the tool uses a combination of image operations (histogram equalization), thresholding on different color spaces (e.g., RGB, YCbCr, Lab, HSV), and binary image processing (boundary and islands removal). As the tool is digitalizing various color scales and classifies pixels into either white (root), green (shoot) or blue (background) categories, the adjustment for white balance is obsolete.”

GWAS was performed separately on traits measured at control, 75mM, and 150mM NaCl treatments. Would it also be informative to map the STI measurement (i.e. plasticity) introduced here?

We thank the reviewer for this important point. We have performed GWAS on both “raw” and STI traits, however, we found that the identified associations were not as abundant as the ones identified with “raw traits”. This makes sense, as we are compounding the root or shoot growth under both conditions, and plastic responses to the environment are expected to be genetically more complex, as they involve more genetic regulators compared to phenotypes that have low plasticity. We have added this as a part of the result description, as we acknowledge that this might be an interesting observation for the field to build upon, and might provide fodder for new methods to deconvolute the complexity in mapping the plastic traits.

“To identify genetic components underlying salt-induced changes in root:shoot ratio, we used the collected data as an input for GWAS. The associations were evaluated based on the p-value, the number of SNPs within the locus, and the number of traits associated with individual loci. As Bonferroni threshold differs depending on the minor allele count (MAC) considered, we identified significant associations based on a Bonferroni threshold for each subpopulation of SNPs based on MAC (Table S3). While we conducted a GWAS on directly measured traits, as well as their Salt Tolerance Index (STI) values, however the amount of associations with STI was much lower compared to directly measured traits (Table S3). This observation aligns with the understanding that plastic responses to environmental conditions tend to be genetically more complex. This complexity likely stems from the involvement of more genetic regulators compared to low-plasticity phenotypes.”

Line 167 - how was LD incorporated into this analysis? Did you use a genome average? Or was LD allowed to vary (as it does) across the genome?

Initially, we have used genome average LD for this purpose (10 kbp for Arabidopsis), and extended the region of interest based on the number of coding genes within the window. We have added this as a part of description to our manuscript.

“For the most promising candidate loci (Table S4), we have identified the gene open reading frames that were located within the genome-wide linkage-disequilibrium (LD) of the associated SNPs. The LD was expanded if multiple SNPs were identified within the region, and the region of interest was expanded based on the number of coding genes within the LD window.”

Line 291 - I think the water potentials are essential, here. What does 50% of soil water holding capacity equal in these soils? In the substrate that we use in our lab, that would represent a considerable soil water deficit even without any salts in the soil.

We thank the reviewer for this comment. As Arabidopsis is occurring naturally in low soil water holding capacity soils (i.e. sandy soils), it is typically growing better in soils that are not very saturated with the water. Throughout many experiments, performed within this study, and other studies performed in our lab (results reported in Awlia et al., 2016 *Frontiers in Plant Science* and Yu & Sussman et al., 2024 *Plant Physiology*), we have not observed any drought like symptoms at 50% soil water holding capacity. The fact that this is reproducible across similar soil types across two laboratories (one in Saudi Arabia and one in the USA) is not to be dismissed. Again - we are currently not equipped to measure water potentials for these plants, as this is not a standard practice (yet) for stress experiments, but we are taking these comments on board for all of our future experiments.

Moreover, our control plants are also “dried down” to 50% of SWHC, and soaked in non-saline water during the “salt stress treatment” to make sure that the soil water saturation is accounted for within the experimental setup. This “dry down” of soil is necessary to ensure equal and effective salt penetration into the soil particles. More details on this method can be found in Awlia et al., 2016.

Again - We have added a new dataset measuring water content in individually soil-grown plants under different conditions as a proxy for soil water status (see new Fig. S24). While we did not observe any significant differences in water content between genotypes under the various conditions, the sr3g mutant showed a slightly higher, though non-significant, water content compared to wild-type Col-0 under control conditions.

We have provided additional information and comments to warn the readers about this method:

“The seeds were germinated in ½ MS media for one week, as described for the agar-based plate experiments. One week after germination, the seedlings were transplanted to the pot (12 x 4 cm insert) containing the Cornell Mix soil (per batch combine: 0.16 m³ of peat moss, 20.84 kg of vermiculite, 0.59 kg of Uni-Mix fertilizer, and 2.27 kg of lime) watered to 100% water holding capacity and placed in the walk-in growth chamber with the 16 h light / 8 h dark period, 22°C and 60% relative humidity throughout the growth period. When all of the pots dried down to the weight corresponding to 50% of their water holding capacity, they were soaked for 1 h in tap water or a 200 mM NaCl solution, resulting in an effective concentration of 100 mM NaCl based on the 50% soil water holding capacity, which corresponded to a moderate level of salt stress (Awlia et al., 2016). The control pots were soaked for the same length of time in 0 mM NaCl solution, to account for the soil saturation effect. We then allowed the pots to be drained for 2-3 h to eliminate excess moisture. The pots were placed under phenotyping rigs equipped with an automated imaging system (Yu et al., 2023) and the pot weight was measured daily to maintain the reference weight corresponding to 50% of the soil water holding capacity throughout the experiment. We would like to note that this gravimetric based method for application of salt stress has been developed for soils typically used for pot-grown plants, with relatively high water holding capacity (Awlia et al. 2016). Within these specific conditions, no drought stress symptoms were observed.”

Lines 415-416 - are these contrasts significant? Figure S3 likewise does not have any notation for significant differences in the means.

We have previously not tested the stronger effect of 125 mM vs 75 mM on relative root and shoot growth, and thus these test results were initially not included in Fig. S3. We have now added the tests and included them within Fig. S3, and added description of their significance into the main body of the manuscript:

“In comparison, the growth rates of the shoot were significantly reduced to 0.71 and 0.43 of the control in 75 and 125 mM NaCl treatments, respectively (Fig. S3). While the mean value of root:shoot growth rate did not change upon salt stress treatment, the variance in the root:shoot ratio significantly expanded with the increasing concentrations of salt (Fig. 1C). These results suggest that while root and shoot growth are well coordinated under non-stress conditions, salt stress exposure results in loss of coordination of organ growth across *Arabidopsis* accessions.”

Line 418 - same comment as preceding. Is this change in variance significant?

We have previously not tested this. We have now added the ANOVA tests and included them within each figure, and added description of their significance into the main body of the manuscript. (see text above)

Line 421 - why would we expect there to be a correlation between root:shoot growth ratio and seedling size?

We were trying to use the seedling size as a proxy for “fitness” - or how well the plants can survive under these specific conditions. We were testing here whether any simple and directional strategy - such as increase or decrease in root:shoot ratio under salt stress - is resulting in better salt tolerance - which would translate into larger overall seedlings. We have rephrased this within the manuscript, to better explain the hypothesis being tested within this specific figure:

“To test whether there is a clear directional correlation between the change in root:shoot ratio and overall salt stress tolerance, we have used the overall seedling size as a proxy for plant salt tolerance (Fig. S4, S5). No significant correlation was found between the root:shoot growth ratio and total seedling size (Fig. S4, S5), indicating that the relationship between coordination of root and shoot growth and salt tolerance during the early seedling establishment is complex.”

Line 438 - I think a stable web link would be more appropriate than listing Dr. Nordborg's email address.

Sorry about this. There is a glitch with our reference citing software. We agree, and thank the reviewer for noticing this! We assigned reference number 43 to it.

Line 439 - I expect that many of your readers may not be experienced with GWAS. Can you provide an explanation as to why only one locus was detected with both the 250K SNP panel and the 4M SNP panel?

We thank the reviewer for raising this point. We have added additional explanation to this observation:

“Increased SNP density can provide more potential associations, highlighting the associated loci with more confidence, due to more SNPs being detected within specific region. The

different panels could capture different LD blocks across the genome. If the locus detected by both panels is in a region of strong LD or under selection, it could be detected consistently. In contrast, other loci may not be captured well by the lower-density 250K SNP panel. The new GWAS revealed 32 additional loci, with only one significantly associated locus being picked up by both 250k and 4M SNPs GWAS (locus 30, Table S3). The detection of only one common locus between the two SNP panels is likely due to differences in resolution, statistical power, and how well each panel captures the genomic regions associated with the trait. ”

| *Figure 2A and B - I suggest adding the p-value cutoff to the y-axis of the Manhattan Plots*

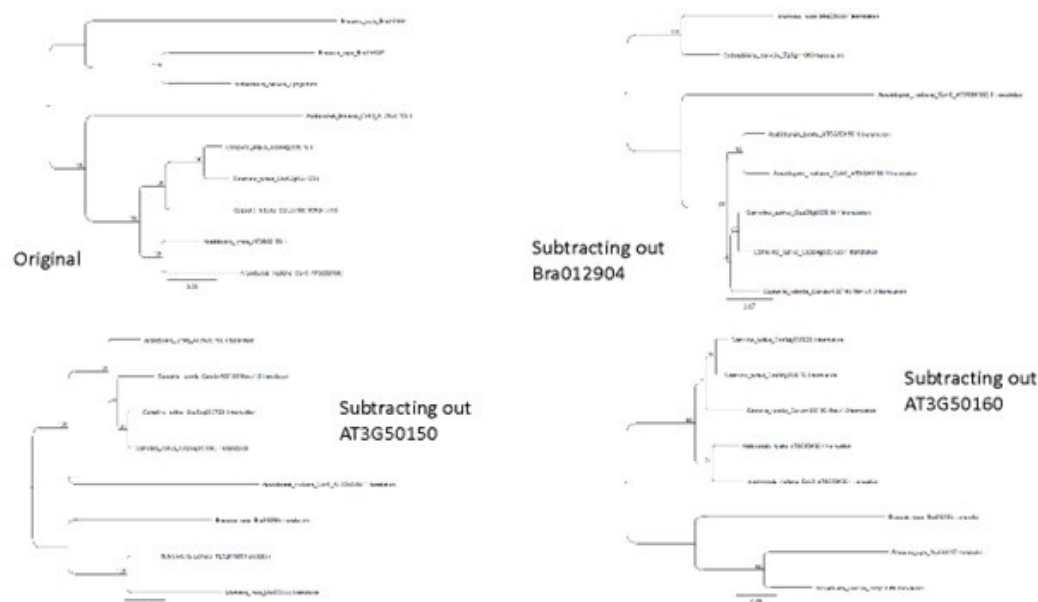
We thank the reviewer for this suggestion, however this is not appropriate. The genome wide p-value cutoffs for GWAS studies are arbitrary, and we have not used a genome-wide cutoff for our SNPs, but rather used cutoffs depending on the minor allele frequency. Therefore, we think adding a straight line to the graphs in Fig. 2A-B representing the overall cutoff, would be misleading. Please see below the text where we explain how the threshold was calculated for individual groups of SNPs with varying MAF:

“The GWAS associations were evaluated for minor allele count (MAC) and association strength above the Bonferroni threshold with $-\log_{10}(p\text{-value}/\#\text{SNPs})$, calculated for each sub-population of SNPs above threshold MAC (Table S3, Bonf.threshold.MAC.specific)”

| *Line 490-492 - Presents the results of the gene tree to support a model in which SR3G diverged from AT3G50150 prior to the speciation events leading to Capsella and Arabidopsis. But this topology requires at least two independent losses of SR3G - can you rule out the hypothesis that the position of SR3G on the gene tree is a result of long branch attraction? Given the syntenic orientation of AT3G50150 and SR3G, and apparent directional selection experienced by the latter lineage, it seems more parsimonious that AT3G50150 and SR3G arose from a very recent duplication event.*

We agree with the reviewer that it seemed most parsimonious for AT3G50160 (SR3G) to be a recent tandem duplication of AT3G50150 – and this was certainly our expectation given the other tandem duplications that have occurred in this genomic region. However, irrespective of the type of alignment from which we built the phylogeny (nucleotide vs AA; sometimes nucleotide is noisier but provides more information) we were never able to recapitulate a tree where AT3G50160 was immediately sister to AT3G50150 – even with a long branch for AT3G50160 indicating a rapid pace of nucleotide/AA change relative to AT3G50150. In regards to long branch attraction, it is our interpretation that long branch attraction typically requires multiple long branches that get placed together at a poorly supported node where sampling is sparse (<https://www.nature.com/articles/s41576-020-0233-0>), whereas we have the single long branch for AT3G50160, and all other A/C clade (Arabidopsis/Camelina/Capsella) members forming a lineage with a much shorter branch. To test the possibility of long branch attraction we subtracted out individual members of the AT3G50150/160 clade to see if there was algorithmic uncertainty in the placement of AT3G50160. We did not observe this in any of the branch subtractions that we performed (see below). Thus, it appears that we must stick with our original interpretation. If the reviewer would like us to soften this interpretation, we would be more than happy to do so, as it does not impact the overall conclusions for AT3G50160 being a rapidly evolving member of this clade.

Author response image 1.



Line 494 (and throughout) - I expect that all of the genes being studied herein are "experiencing selection," even if it's boring-old purifying selection on functionally conserved proteins. I think you mean to say "directional selection."

We thank the reviewer for this comment and completely agree that we lacked precision on our statement. We have corrected this throughout the manuscript.

Line 497 - state the background and foreground values of omega, here.

We apologize for not including these values and have added them at this point in the manuscript (new Table S6).

Line 511 and Line 673 - Inspection of Figure S13B suggests that SR3G is not "predominantly" expressed nor does it have the "highest enrichment" in the root stele. Certainly, among root cell types, this is predominant. But it appears to be quite highly expressed in late-stage seeds and some floral organs, as well.

We appreciate the reviewer for recognizing that SR3G is not a highly expressed gene. In root cell types, its expression is enriched in the root stele. Overall, SR3G is expressed at both early and later developmental stages. Our investigation of later developmental stages related to seed production did not reveal any significant phenotypic differences in fertility.

Line 514 - "54-folds" should be "54-fold."

Thanks. We made corrections.

Figure 7 - For symmetry, I suggest adding the "Beginning of salt stress" arrow to the "Early Stress" panel as well (even if it's right at day 0).

Thanks. We added the arrow to Early Stress in both Panels A and B.

| *Figure S2 - both graphs should have the same scale on the y-axis*

Thanks - we have now re-plotted the graph with the matching y-axis scales.

| *Line 531 - I feel that this is a significant overstatement. The strongest statement supported by the results presented here is that SR3G is the most prominent DUF247 studied herein in root development under salt stress.*

Thanks for the comments. We rephrase the statement.

“These results suggest that SR3G is the most prominent DUF247 studied within our study to affect root development under salt stress.”

| *Lines 583-605 - These data seem to me to be tangential to the central aims of the study. I suggest removing them for clarity/brevity.*

We greatly appreciate the reviewer's suggestion. Our study primarily focused on characterizing the main GWAS candidate, SR3G. Since SR3G is located within a cluster of other DUF247 genes on chromosome 3, we believe that screening the neighboring DUF247 genes could provide further insights into SR3G's role in root development. Additionally, we believe that the generated data and lines will serve as a valuable resource for other researchers interested in studying these genes. For these reasons, we have decided to retain these datasets in the manuscript.

| *Lines 650-652 - these sections 1-3 differences in suberization between SR3G and Col-0 under control conditions are not significant. At best, this may be described as a "trend" and not "higher levels." In section 4, it is VERY marginally significant (and probably not at all after the large number of tests performed, here.)*

We appreciate the reviewer's feedback and have revised the wording accordingly.

| *Line 660 - this statement is only true for Section 1. I suggest adding this caveat.*

We appreciate the reviewer's comments on this matter. We quantified four suberin monomers in whole root seedlings rather than in individual root sections due to the technical challenges of separating the sections without microscopy and the limited availability of samples for GS-MS analysis.

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